

Transmission lines:

General Considerations

By modeling transmission lines in the form of equivalent circuits, we can use Kirchhoff's voltage and current laws to develop wave equations whose solutions provide an understanding of wave propagation, standing waves, and power transfer.

Although the notion of transmission lines may encompass all structures and media that serve to transfer energy or information between two points, including wave fibers in the human body and fluid and solids that support the propagation of mechanical pressure waves, this chapter focuses on transmission lines that guide electromagnetic signals. Such transmission lines include telephone wires, coaxial cables carrying audio and video information to TV sets or digital data to computer monitors, microstrip printed on microwave circuit boards, and optical fibers carrying light waves for the transmission of data at very high rates. Fundamentally, a transmission line is a two-port network,

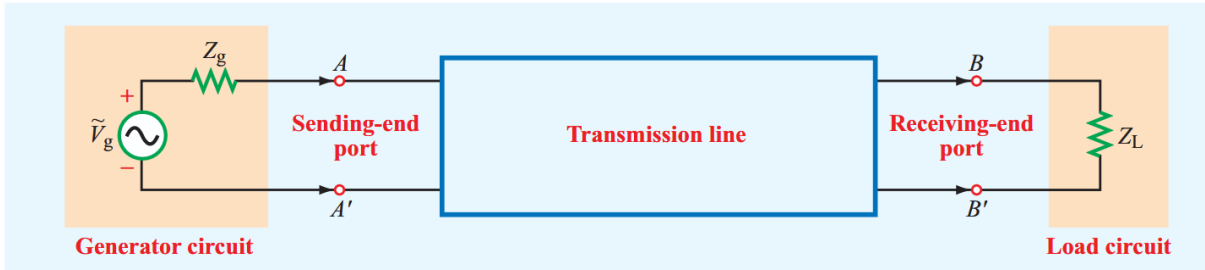


Figure 1-1: A transmission line is a two-port network connecting a generator circuit at the sending end to a load at the receiving end

with each port consisting of two terminals, as illustrated in Fig.1.1. One of the ports, the line's sending end, is connected to a source (also called the generator). The other port, the line's receiving end, is connected to a load. The source connected to the transmission line's sending end may be any circuit generating an output voltage, such as a radar transmitter, an amplifier, or a computer terminal operating in transmission mode. From circuit theory, a dc source can be represented by a Thévenin-equivalent generator circuit consisting of a generator voltage V_g in series with a generator resistance R_g , as shown in Fig.1.1. In the case of alternating-current (ac) signals, the generator circuit is represented by a voltage phasor $\tilde{V}_g$ and an impedance Z_g .

The load circuit, or simply the load, may be a transmitting antenna in the case of radar, a computer terminal operating in the receiving mode, the input terminals of an amplifier, or any output circuit whose input terminals can be represented by an equivalent load impedance Z_L .

The Role of Wavelength

In low-frequency circuits, circuit elements usually are interconnected using simple wires. In the circuit shown in Fig.1.2, for example, the generator is connected to a simple RC load via a pair of wires. In view of our definition in the preceding paragraphs of what constitutes a transmission line, we pose the question: Is the pair of wires between terminals AA' and terminals BB' a transmission line? If so, under what set of circumstances should we explicitly treat the pair of wires as a transmission line as opposed to ignoring

their presence altogether and treating the circuit as only an RC load connected to a generator $\tilde{V}_g$? The answer to the first question is: Yes. The pair of wires does constitute a transmission line. The answer to the second question is: The factors that determine whether or not we should treat the wires as a transmission line is governed by the length of the line l and the frequency f of the signal provided by the generator. (As we will see later, the determining factor is not the length l to wavelength λ of the wave propagating on the transmission line between the source at terminals AA' and the load at terminals BB' .)

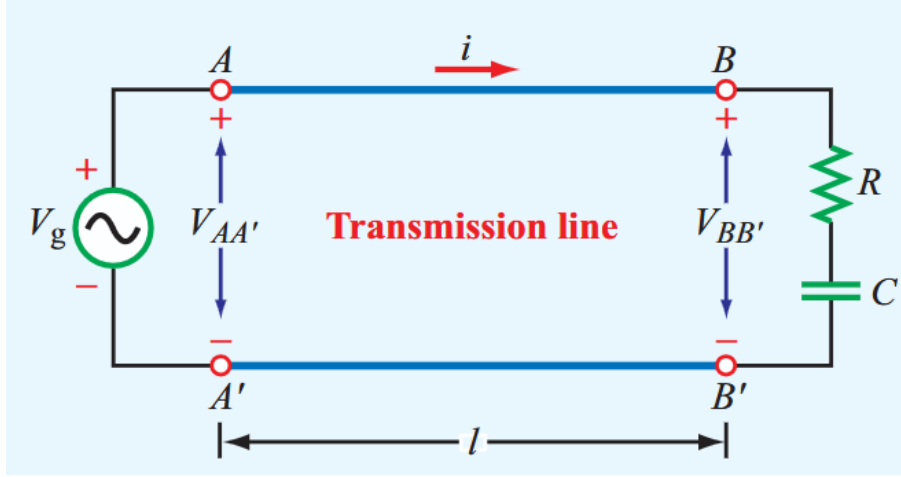


Figure 1-2: Generator connected to an RC circuit through a transmission line of length l .

If the generator voltage is cosinusoidal in time, then the voltage across the input terminals AA' is

$$V_{AA'} = V_g(t) = V_0 \cos \omega t \quad (\text{V}), \quad (1-1)$$

where $\omega = 2\pi f$ is the angular frequency. If we assume that the current flowing through the wires travels at the speed of light, $c = 3 \times 10^8$ m/s, then the voltage across the output terminals BB' will have to be delayed in time relative to that across AA' by the travel delay time l/c . Thus, assuming no ohmic losses in the transmission line and ignoring other transmission-line effects discussed later in this chapter,

$$V_{BB'}(t) = V_{AA'}(t - l/c) = V_0 \cos[\omega(t - l/c)] = V_0 \cos(\omega t - \phi_0), \quad (1-2)$$

with

$$\phi_0 = \frac{\omega l}{c} \quad (\text{rad}). \quad (1-3)$$

Thus, the time delay associated with the length of the line manifests itself as a constant phase shift ϕ_0 in the argument of the cosine. Let us compare $V_{BB'}$ with $V_{AA'}$ at $t = 0$ for an ultralow-frequency electronic circuit operating at a frequency $f = 1$ kHz. For a typical wire length $l = 5$ cm, Eqs. (1-1) and (1-2) give $V_{AA'} = V_0$ and $V_{BB'} = V_0 \cos(2\pi f l/c) = 0.9999999908V_0$. Hence, for all practical purposes, the presence of the transmission line may be ignored, and terminal AA' and terminal BB' are treated as identical to $V_{BB'} = 0.91V_0$, which is a deviation of 9%. Furthermore, had the frequency of the signal on the 5-cm long wire been 1.5 GHz instead of 1 kHz, the voltage at the end of the line would

have been $V_{BB'} = 0$! Thus, the pair of wires would have a voltage V_0 at AA' and zero at BB' simultaneously. The determining factor is the magnitude of $\phi_0 = \omega l/c$. From Eq. (1-3), the velocity of propagation u_p of a traveling wave is related to the oscillation frequency f and the wavelength λ by

$$u_p = f\lambda \quad (\text{m/s}). \quad (1-4)$$

In the present case, $u_p = c$. Hence, the phase delay is

$$\phi_0 = \frac{\omega l}{c} = \frac{2\pi f l}{c} = 2\pi \frac{l}{\lambda} \quad \text{radians}. \quad (1-5)$$

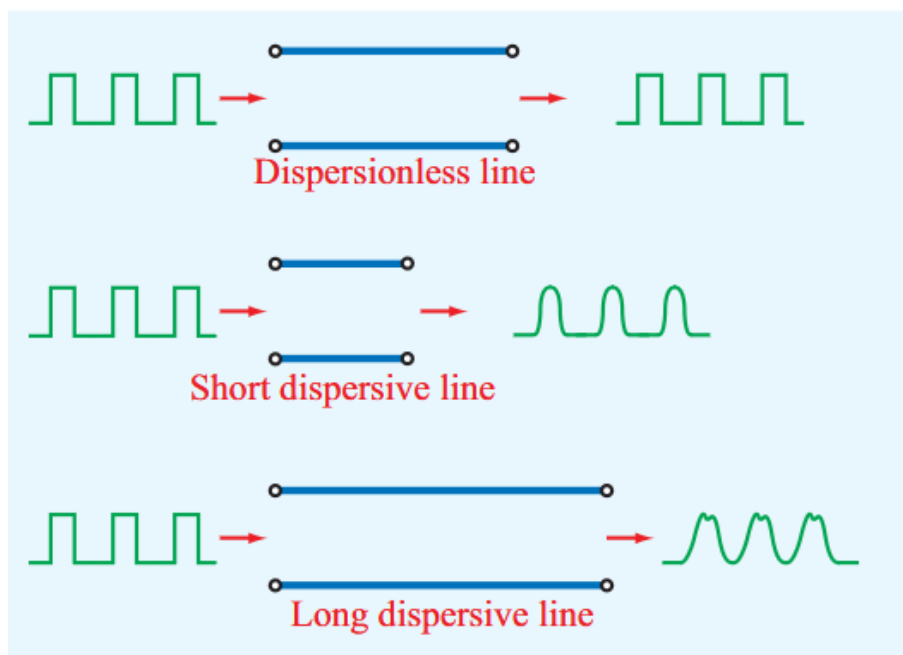


Figure 1-3: A dispersionless line does not distort signals passing through it regardless of its length, whereas a dispersive line distorts the shape of the input pulses because the different frequency components propagate at different velocities. The degree of distortion is proportional to the length of the dispersive line.

► When l/λ is very small, transmission-line effects may be ignored, but when $l/\lambda \gtrsim 0.01$, it may be necessary to account not only for the phase shift due to the time delay but also for the presence of reflected signals that may have been bounced back by the load toward the generator.

Power loss on the line and dispersive effects may need to be considered as well.

► A dispersive transmission line is one on which the wave velocity is not constant as a function of the frequency f .

This means that the shape of a rectangular pulse, which through Fourier analysis can be decomposed into many sinusoidal waves of different frequencies, gets distorted as it travels down the line because its different frequency components do not propagate at the same velocity (Fig.1.3). Preservation of pulse shape is very important in high-speed

data transmission not only between terminals but also across transmission line segments fabricated within high-speed integrated circuits. At 10 GHz, for example, the wavelength is $\lambda = 3$ cm in air but only on the order of 1 cm in a semiconductor material. Hence, even lengths between devices on the order of millimeters become significant, and their presence has to be accounted for in the design of the circuit.

Lumped-Element Model

When we draw a schematic of an electronic circuit, we use specific symbols to represent resistors, capacitors, inductors, diodes, and the like. In each case, the symbol represents the functionality of the device rather than its shape, size, or other attributes. We shall do the same for transmission lines.

► A transmission line will be represented by a parallel-wire configuration (Fig. 1-4(a)), regardless of its specific shape or constitutive parameters.

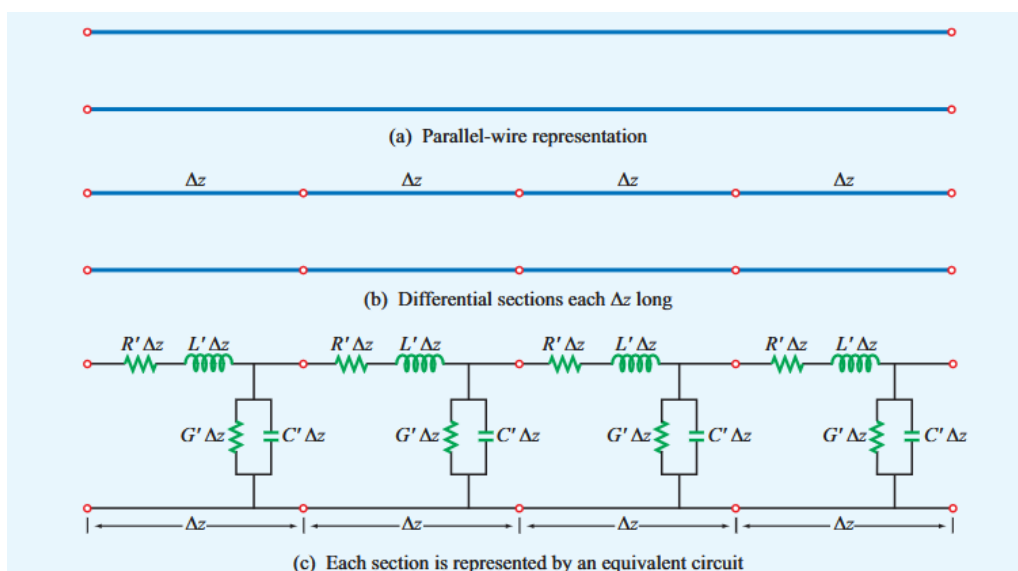


Figure 1-4: Regardless of its cross-sectional shape, a TEM transmission line is represented by the parallel-wire configuration shown in part (a). To obtain equations relating voltages and currents, the line is subdivided into small differential sections in part (b), each of which is then represented by an equivalent circuit in part (c)

Thus, Fig. 1-4(a) may represent a coaxial line, a two-wire line, or any other TEM line.

Drawing again on our familiarity with electronic circuits, when we analyze a circuit containing a transistor, we mimic the functionality of the transistor by an equivalent circuit composed of sources, resistors, and capacitors. We apply the same approach to the transmission line by orienting the line along the z direction, subdividing it into differential sections each of length Δz (Fig. 1.4(b)), and then representing each section by an equivalent circuit, as illustrated in Fig. 1.4(c). This representation, often called the *lumped-element circuit* model, consists of four basic elements with values that henceforth will be called the *transmission line parameters*. These are:

- R' : The combined *resistance* of both conductors per unit length, in Ω/m ,

- L' : The combined *inductance* of both conductors per unit length, in H/m,
- G' : The *conductance* of the insulation medium between the two conductors per unit length, in S/m, and
- C' : The *capacitance* of the two conductors per unit length, in F/m.

Whereas the four line parameters are characterized by different formulas for different types of transmission lines, the equivalent model represented by Fig. 1.4(c) is equally applicable to all TEM transmission lines. The prime superscript is used as a reminder that the line parameters are differential quantities whose units are per unit length.

Transmission-Line Equations:

A transmission line usually connects a source on one end to a load on the other. Before considering the complete circuit, however, we will develop general equations that describe the voltage across and current carried by the transmission line as a function of time t and spatial position z . Using the lumped-element model of Fig. 1.4(c), we begin by considering a differential length Δz as shown in Fig. 1.5. The quantities $v(z, t)$ and $i(z, t)$

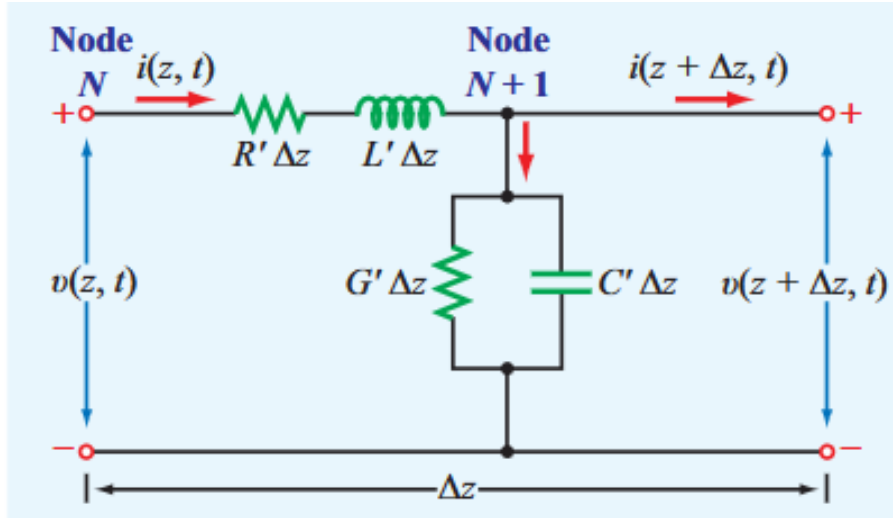


Figure 1-5: Equivalent circuit of a two-conductor transmission line of differential length Δz .

denote the instantaneous voltage and current at the left end of the differential section (node N), and similarly $v(z + \Delta z, t)$ and $i(z + \Delta z, t)$ denote the same quantities at node $(N + 1)$, located at the right end of the section. Application of Kirchhoff's voltage law accounts for the voltage drop across the series resistance $R'\Delta z$ and inductance $L'\Delta z$:

$$v(z, t) - R'\Delta z i(z, t) - L'\Delta z \frac{\partial i(z, t)}{\partial t} - v(z + \Delta z, t) = 0. \quad (1-6)$$

Upon dividing all terms by Δz and rearranging them, we obtain

$$-\left[\frac{v(z + \Delta z, t) - v(z, t)}{\Delta z} \right] = R' i(z, t) + L' \frac{\partial i(z, t)}{\partial t}. \quad (1-7)$$

In the limit as $\Delta z \rightarrow 0$, Eq. (1-7) becomes a differential equation:

$$-\frac{\partial v(z, t)}{\partial z} = R'i(z, t) + L'\frac{\partial i(z, t)}{\partial t}. \quad (1-8)$$

Similarly, Kirchhoff's current law accounts for current drawn from the upper line at node $(N + 1)$ by the parallel conductance $G'\Delta z$ and capacitance $C'\Delta z$:

$$i(z, t) - G'\Delta z v(z + \Delta z, t) - C'\Delta z \frac{\partial v(z + \Delta z, t)}{\partial t} - i(z + \Delta z, t) = 0. \quad (1-9)$$

Upon dividing all terms by Δz and taking the limit $\Delta z \rightarrow 0$, Eq. (1-9) becomes a second-order differential equation:

$$-\frac{\partial i(z, t)}{\partial z} = G'v(z, t) + C'\frac{\partial v(z, t)}{\partial t}. \quad (1-10)$$

The first-order differential equations (1-8) and (1-10) are the time-domain forms of the *transmission-line equations*, known as the *telegrapher's equations*.

We make use of the phasor representation with a cosine reference. Thus, we define

$$v(z, t) = \Re e \left[\tilde{V}(z)e^{j\omega t} \right], \quad (1-11)$$

$$i(z, t) = \Re e \left[\tilde{I}(z)e^{j\omega t} \right], \quad (1-12)$$

where $\tilde{V}(z)$ and $\tilde{I}(z)$ are the phasor counterparts of $v(z, t)$ and $i(z, t)$, respectively, each of which may be real or complex. Upon substituting Eqs. (1-11) and (1-12) into Eqs. (1-8) and (1-10), and utilizing the property given that $\partial/\partial t$ in the time domain is equivalent to multiplication by $j\omega$ in the phasor domain, we obtain the following pair of equations:

$$-\frac{d\tilde{V}(z)}{dz} = (R' + j\omega L')\tilde{I}(z), \quad (1-13)$$

$$-\frac{d\tilde{I}(z)}{dz} = (G' + j\omega C')\tilde{V}(z). \quad (1-14)$$

(telegrapher's equations in phasor form)

Wave Propagation on a Transmission Line

The two first-order coupled equations (1-13) and (1-14) can be combined to give two second-order uncoupled wave equations: one for $\tilde{V}(z)$ and another for $\tilde{I}(z)$. The wave equation for $\tilde{V}(z)$ is derived by first differentiating both sides of Eq. (1-13) with respect to z , resulting in

$$-\frac{d^2\tilde{V}(z)}{dz^2} = (R' + j\omega L')\frac{d\tilde{I}(z)}{dz}. \quad (1-15)$$

Then, upon substituting Eq. (1-14) for $d\tilde{I}(z)/dz$, Eq. (1-15) becomes

$$\frac{d^2 \tilde{V}(z)}{dz^2} - (R' + j\omega L')(G' + j\omega C')\tilde{V}(z) = 0, \quad (1-16)$$

or

$$\frac{d^2 \tilde{V}(z)}{dz^2} - \gamma^2 \tilde{V}(z) = 0, \quad (1-17)$$

(wave equation for $\tilde{V}(z)$)

where

$$\gamma = \sqrt{(R' + j\omega L')(G' + j\omega C')}. \quad (1-18)$$

(propagation constant)

Application of the same steps to Eqs. (1-13) and (1-14) in reverse order leads to

$$\frac{d^2 \tilde{I}(z)}{dz^2} - \gamma^2 \tilde{I}(z) = 0. \quad (1-19)$$

(wave equation for $\tilde{I}(z)$)

The second-order differential equations (1-17) and (1-19) are called *wave equations* for $\tilde{V}(z)$ and $\tilde{I}(z)$, respectively, and γ is called the *complex propagation constant* of the transmission line. As such, γ consists of a real part α , called the *attenuation constant* of the line with units of Np/m, and an imaginary part β , called the *phase constant* of the line with units of rad/m. Thus,

$$\gamma = \alpha + j\beta \quad (1-20)$$

with

$$\alpha = \Re(\gamma) = \Re\left(\sqrt{(R' + j\omega L')(G' + j\omega C')}\right) \quad (\text{Np/m}), \quad (1-21)$$

(attenuation constant)

$$\beta = \Im(\gamma) = \Im\left(\sqrt{(R' + j\omega L')(G' + j\omega C')}\right) \quad (\text{rad/m}). \quad (1-22)$$

(phase constant)

Phasor-Domain Solution

In Eqs. (1-21) and (1-22), we choose the square-root solutions that give positive values for α and β . For passive transmission lines, α is either zero or positive. Most transmission lines, and all those considered in this chapter, are of the passive type. The gain region of a laser is an example of an active transmission line with a negative α .

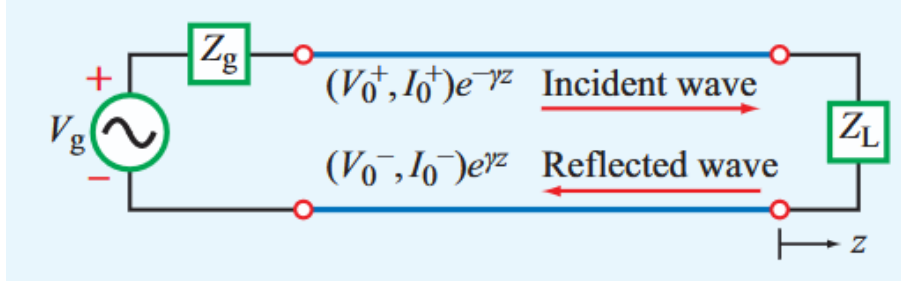


Figure 1-6: In general, a transmission line can support two traveling waves: an incident wave (with voltage and current amplitudes (V_0^+, I_0^+)) traveling along the $+z$ direction (towards the load) and a reflected wave (with (V_0^-, I_0^-)) traveling along the $-z$ direction (towards the source).

The wave equations (1-17) and (1-19) have traveling wave solutions of the forms

$$\tilde{V}(z) = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z} \quad (\text{V}), \quad (1-23)$$

$$\tilde{I}(z) = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z} \quad (\text{A}). \quad (1-24)$$

As shown later, the $e^{-\gamma z}$ term represents a wave propagating in the $+z$ direction, while the $e^{\gamma z}$ term represents a wave propagating in the $-z$ direction (Fig. 1.6). Verification that these are indeed valid solutions is easily accomplished by substituting the expressions given by Eqs. (1-23) and (1-24), as well as their second derivatives, into Eqs. (1-17) and (1-19).

In their present form, the solutions given by Eqs. (1-23) and (1-24) contain four unknowns: the wave amplitudes (V_0^+, I_0^+) of the $+z$ propagating wave and (V_0^-, I_0^-) of the $-z$ propagating wave. We can easily relate the current amplitudes, I_0^+ and I_0^- , to the voltage wave amplitudes, V_0^+ and V_0^- , by using Eq. (1-13) in Eq. (1-23) and then solving for the current $\tilde{I}(z)$. The process leads to

$$\tilde{I}(z) = \frac{\gamma}{R' + j\omega L'} [V_0^+ e^{-\gamma z} - V_0^- e^{\gamma z}]. \quad (1-25)$$

Comparison of each term with the corresponding term in Eq. (1-24) leads us to conclude that

$$\frac{V_0^+}{I_0^+} = Z_0 = -\frac{V_0^-}{I_0^-}, \quad (1-26)$$

where

$$Z_0 = \frac{R' + j\omega L'}{\gamma} = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}} \quad (\Omega), \quad (1-27)$$

is called the *characteristic impedance* of the line.

► It should be noted that Z_0 is equal to the ratio of the voltage amplitude to the current amplitude for each of the traveling waves individually (with an additional minus sign in the case of the $-z$ propagating wave), but it is not equal to the ratio of the total voltage $\tilde{V}(z)$ to the total current $\tilde{I}(z)$ unless one of the two waves is absent.

It seems reasonable that the voltage-to-current ratios of the two waves V_0^+/I_0^+ and V_0^-/I_0^- are both related to the same quantity, namely Z_0 , but it is not immediately obvious as to why one of the ratios is the negative of the other. On a transmission line, the voltage is related to the electric field $\mathbf{E}$ and the current is related to the magnetic field $\mathbf{H}$. To satisfy the directional rule, reversing the direction of propagation requires reversal of the direction (or polarity) of I relative to V . Hence,

$$\frac{V_0^-}{I_0^-} = -\frac{V_0^+}{I_0^+}.$$

In terms of Z_0 , Eq. (1-25) can be cast in the form

$$\tilde{I}(z) = \frac{V_0^+}{Z_0}e^{-\gamma z} - \frac{V_0^-}{Z_0}e^{\gamma z}. \quad (1-28)$$

According to Eq. (1-27), the characteristic impedance Z_0 is determined by the angular frequency ω of the wave traveling along the line and the four line parameters (R', L', G', C'). These, in turn, are determined by the geometry and its constitutive parameters. Consequently, the combination of Eqs. (1-23) and (1-28) now contains only two unknowns, namely V_0^+ and V_0^- , as opposed to four.

Time-Domain Solution for $v(z, t)$

In later sections, we apply boundary conditions at the source and load ends of the transmission line to obtain expressions for the remaining wave amplitudes V_0^+ and V_0^- . In general, each is a complex quantity characterized by a magnitude and a phase angle:

$$V_0^+ = |V_0^+|e^{j\phi^+}, \quad (1-29)$$

$$V_0^- = |V_0^-|e^{j\phi^-}. \quad (1-30)$$

After substituting these definitions in Eq. (1-23) and using Eq. (1-20) to decompose γ into its real and imaginary parts, we can convert back to the time domain to obtain an expression for $v(z, t)$, the instantaneous voltage on the line:

$$\begin{aligned} v(z, t) &= \Re \left[\tilde{V}(z)e^{j\omega t} \right] \\ &= \Re \left[(V_0^+e^{-\gamma z} + V_0^-e^{\gamma z})e^{j\omega t} \right] \\ &= \Re \left[|V_0^+|e^{j\phi^+}e^{-(\alpha+j\beta)z} + |V_0^-|e^{j\phi^-}e^{(\alpha+j\beta)z} \right]. \end{aligned}$$

The final expression for $v(z, t)$ is then given by

$$v(z, t) = |V_0^+|e^{-\alpha z} \cos(\omega t - \beta z + \phi^+) + |V_0^-|e^{\alpha z} \cos(\omega t + \beta z + \phi^-). \quad (1-31)$$

We recognize the first term in Eq. (1-31) as a wave traveling in the $+z$ direction (the coefficients of t and z have opposite signs) and the second term as a wave traveling in the $-z$ direction (the coefficients of t and z are both positive). Both waves propagate with a phase velocity u_p given by :

$$u_p = f\lambda = \frac{\omega}{\beta}. \quad (1-32)$$

Because the wave is guided by the transmission line, λ often is called the *guide wavelength*. The factor $e^{-\alpha z}$ accounts for the attenuation of the $+z$ -propagating wave, and the factor $e^{\alpha z}$ accounts for the attenuation of the $-z$ -propagating wave.

► The presence of two waves on the line propagating in opposite directions produces a *standing wave*.

Time-Domain Solution for $i(z, t)$

In the phasor domain, the expression for $\tilde{I}(z)$ given by Eq. (1-28) is similar to that for $\tilde{V}(z)$, except that the current amplitudes are divided by the characteristic impedance of the transmission line, Z_0 , and the current wave traveling in the $-z$ direction has a minus sign (second term in Eq. (1-28)). Since Z_0 is, in general, a complex quantity [see Eq. (1-27)], we need to express it as

$$Z_0 = |Z_0|e^{j\phi_z}. \quad (1-33)$$

To obtain an expression for $i(z, t)$, we insert the expressions given by Eqs. (1-29) and (1-33) into Eq. (1-28), multiply both terms by $e^{j\omega t}$, and then take the real part:

$$\begin{aligned} i(z, t) &= \Re \left[\tilde{I}(z)e^{j\omega t} \right] \\ &= \Re \left[\frac{|V_0^+|}{|Z_0|} e^{j\phi^+} e^{-j\phi_z} e^{-\alpha z} e^{j(\omega t - \beta z)} \right. \\ &\quad \left. - \frac{|V_0^-|}{|Z_0|} e^{j\phi^-} e^{-j\phi_z} e^{\alpha z} e^{j(\omega t + \beta z)} \right], \end{aligned}$$

which yields

$$i(z, t) = \frac{|V_0^+|}{|Z_0|} e^{-\alpha z} \cos(\omega t - \beta z + \phi^+ - \phi_z) - \frac{|V_0^-|}{|Z_0|} e^{\alpha z} \cos(\omega t + \beta z + \phi^- - \phi_z). \quad (1-34)$$

Comparison of the expression for $i(z, t)$ with that given by Eq. (1-31) for $v(z, t)$ reveals that:

- (a) The amplitudes of the current waves are reduced by $|Z_0|$ when compared with the amplitudes of the voltage waves.
- (b) The amplitude of the second term in Eq. (1-34), corresponding to the current wave traveling in the negative z direction, has a minus sign.
- (c) The reference phase angle in the expression for $i(z, t)$ has an extra component, namely $-\phi_z$, when compared with the reference phase angle of $v(z, t)$.

The Lossless Transmission Line: General Considerations

According to the preceding section, a transmission line is fully characterized by two fundamental parameters: its propagation constant γ and its characteristic impedance Z_0 , both of which are specified by the angular frequency ω and the line parameters R' , L' , G' , and C' .

► In many practical situations, the transmission line can be designed to exhibit low ohmic losses by selecting conductors with very high conductivities and dielectric materials (separating the conductors) with negligible conductivities. As a result, R' and G' assume very small values such that $R' \ll \omega L'$ and $G' \ll \omega C'$.

These conditions allow us to set $R' = G' \approx 0$ in Eq. (1-18), which yields

$$\gamma = \alpha + j\beta = j\omega\sqrt{L'C'}, \quad (1-35)$$

which in turn implies that

$$\alpha = 0 \quad (\text{lossless line}), \quad (1-36)$$

$$\beta = \omega\sqrt{L'C'} \quad (\text{lossless line}). \quad (1-37)$$

For the characteristic impedance, application of the lossless line conditions to Eq. (1-27) leads to

$$Z_0 = \sqrt{\frac{L'}{C'}} \quad (\text{lossless line}), \quad (1-38)$$

which now is a real number. Using the lossless line expression for β [Eq. (1-37)], we obtain the following expressions for the guide wavelength λ and the phase velocity u_p :

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{\omega\sqrt{L'C'}}, \quad (1-39)$$

$$u_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{L'C'}}. \quad (1-40)$$

Upon using

$$L'C' = \mu\epsilon$$

Eqs. (1-37) and (1-40) may be rewritten as

$$\beta = \omega\sqrt{\mu\epsilon} \quad (\text{rad/m}), \quad (1-41)$$

$$u_p = \frac{1}{\sqrt{\mu\epsilon}} \quad (\text{m/s}), \quad (1-42)$$

where μ and ϵ are, respectively, the magnetic permeability and electric permittivity of the insulating material separating the conductors. Materials used for this purpose are usually characterized by a permeability $\mu_0 = 4\pi \times 10^{-7}$ H/m (the permeability of free space). Also, the permittivity is often specified in terms of the relative permittivity ϵ_r defined as

$$\epsilon_r = \epsilon/\epsilon_0, \quad (1-43)$$

where $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m} \approx (1/36\pi) \times 10^{-9} \text{ F/m}$ is the permittivity of free space (vacuum). Hence, Eq. (1-42) becomes

$$u_p = \frac{1}{\sqrt{\mu_0 \epsilon_r \epsilon_0}} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \frac{1}{\sqrt{\epsilon_r}} = \frac{c}{\sqrt{\epsilon_r}}, \quad (1-44)$$

where $c = 1/\sqrt{\mu_0 \epsilon_0} = 3 \times 10^8 \text{ m/s}$ is the velocity of light in free space. If the insulating material between the conductors is air, then $\epsilon_r = 1$ and $u_p = c$. In view of Eq. (1-43), the relationship between λ and u_p given by Eq. (1-32), the wavelength is given by

$$\lambda = \frac{u_p}{f} = \frac{c}{f} \frac{1}{\sqrt{\epsilon_r}} = \frac{\lambda_0}{\sqrt{\epsilon_r}}, \quad (1-45)$$

where $\lambda_0 = c/f$ is the wavelength in air corresponding to a frequency f . Note that, because both u_p and λ depend on ϵ_r , the choice of the type of insulating material used in a transmission line is dictated not only by its mechanical properties but by its electrical properties as well.

According to Eq. (1-44), if ϵ_r of the insulating material is independent of f (which usually is the case for commonly used TEM lines), the same independence applies to u_p .

► If sinusoidal waves of different frequencies travel on a transmission line with the same phase velocity, the line is called *nondispersive*.

This is an important feature to consider when digital data are transmitted in the form of pulses. A rectangular pulse or a series of pulses is composed of many Fourier components with different frequencies. If the phase velocity is the same for all frequency components (or at least for the dominant ones), then the pulse's shape does not change as it travels down the line. In contrast, the shape of a pulse propagating in a dispersive medium becomes progressively distorted, and the pulse length increases (stretches out) as a function of the distance traveled in the medium (Fig. 1-3), thereby imposing a limitation on the maximum data rate (which is related to the length of the individual pulses and the spacing between adjacent pulses) that can be transmitted through the medium without loss of information.

Voltage Reflection Coefficient

With $\gamma = j\beta$ for the lossless line, Eqs. (1-23) and (1-28) for the total voltage and current become

$$\tilde{V}(z) = V_0^+ e^{-j\beta z} + V_0^- e^{j\beta z}, \quad (1-46)$$

$$\tilde{I}(z) = \frac{V_0^+}{Z_0} e^{-j\beta z} - \frac{V_0^-}{Z_0} e^{j\beta z}. \quad (1-47)$$

These expressions contain two unknowns, V_0^+ and V_0^- . An exponential factor of the form $e^{-j\beta z}$ is associated with a wave traveling in the positive z direction, from the source (sending end) to the load (receiving end). Accordingly, we refer to it as the *incident wave* with V_0^+ as its voltage amplitude. Similarly, the term containing $V_0^- e^{j\beta z}$ represents a

reflected wave with voltage amplitude V_0^- traveling along the negative z direction from the load to the source.

To determine V_0^+ and V_0^- , we need to consider the lossless transmission line in the context of the complete circuit, including a generator circuit at its input terminals and a load at its output terminals, as shown in Fig. 1.7. The line of length l is terminated in an arbitrary load impedance Z_L .

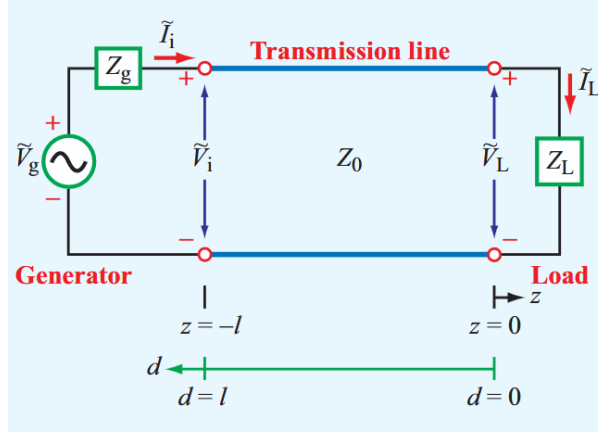


Figure 1-7: Transmission line of length l connected on one end to a generator circuit and on the other end to a load Z_L . The load is located at $z = 0$, and the generator terminals are at $z = -l$. Coordinate d is defined as $d = -z$.

► For mathematical convenience, the reference of the spatial coordinate z is chosen such that $z = 0$ corresponds to the location of the load, not the generator.

At the sending end, at $z = -l$, the line is connected to a sinusoidal voltage source with phasor voltage $\tilde{V}_g$ and internal impedance Z_g . Since z points from the generator to the load, positive values of z correspond to locations beyond the load; therefore, they are irrelevant to our circuit. In future sections, we will find it more convenient to work with a spatial dimension that also starts at the load but whose direction is opposite of z . We shall call it the *distance from the load* d and define it as $d = -z$, as shown in Fig. 1.7.

The phasor voltage across the load, $\tilde{V}_L$, and the phasor current through it, $\tilde{I}_L$, are related by the load impedance Z_L ,

$$Z_L = \frac{\tilde{V}_L}{\tilde{I}_L}. \quad (1-48)$$

The voltage $\tilde{V}_L$ is the total voltage on the line $\tilde{V}(z)$ given by Eq. (1-46), and $\tilde{I}_L$ is the total current $\tilde{I}(z)$ given by Eq. (1-47). Both are evaluated at $z = 0$:

$$\tilde{V}_L = \tilde{V}(z = 0) = V_0^+ + V_0^-, \quad (1-49)$$

$$\tilde{I}_L = \tilde{I}(z = 0) = \frac{V_0^+}{Z_0} - \frac{V_0^-}{Z_0}. \quad (1-50)$$

Using these expressions in Eq. (1-48), we obtain

$$Z_L = \left(\frac{V_0^+ + V_0^-}{V_0^+ - V_0^-} \right) Z_0. \quad (1-51)$$

Solving for V_0^- gives

$$V_0^- = \left(\frac{Z_L - Z_0}{Z_L + Z_0} \right) V_0^+. \quad (1-52)$$

► The ratio of the amplitudes of the reflected and incident voltage waves at the load is known as the *voltage reflection coefficient* Γ .

From Eq. (1-52), it follows that

$$\Gamma = \frac{V_0^-}{V_0^+} = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{Z_L/Z_0 - 1}{Z_L/Z_0 + 1} = \frac{z_L - 1}{z_L + 1} \quad (\text{dimensionless}), \quad (1-53)$$

where

$$z_L = \frac{Z_L}{Z_0} \quad (1-54)$$

is the *normalized load impedance*. In many transmission line problems, we can streamline the necessary computation by normalizing all impedances in the circuit to the characteristic impedance Z_0 . Normalized impedances are denoted by lower-case letters.

In view of Eq. (1-26), the ratio of the current amplitudes is

$$\frac{I_0^-}{I_0^+} = -\frac{V_0^-}{V_0^+} = -\Gamma. \quad (1-55)$$

► We note that whereas the ratio of the voltage amplitudes is equal to Γ , the ratio of the current amplitudes is equal to $-\Gamma$.

The reflection coefficient Γ is governed by a single parameter, the normalized load impedance z_L . As indicated by Eq. (1-38), Z_0 of a lossless line is a real number. However, Z_L is in general a complex quantity, as in the case of a series RL circuit, for example, for which $Z_L = R + j\omega L$. Hence, in general Γ also is complex and given by

$$\Gamma = |\Gamma|e^{j\theta_r}, \quad (1-56)$$

where $|\Gamma|$ is the magnitude of Γ and θ_r is its phase angle. Note that $|\Gamma| \leq 1$.

► A load is said to be *matched* to a transmission line if $Z_L = Z_0$ because then there will be no reflection by the load ($\Gamma = 0$ and $V_0^- = 0$).

On the other hand, when the load is an open circuit ($Z_L = \infty$), $\Gamma = 1$ and $V_0^- = V_0^+$, and when it is a short circuit ($Z_L = 0$), $\Gamma = -1$ and $V_0^- = -V_0^+$ (Fig 1.8).

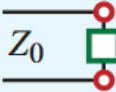
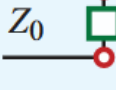
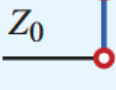
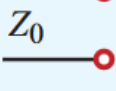
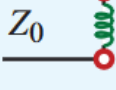
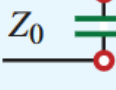
Reflection Coefficient $\Gamma = \Gamma e^{j\theta_r}$		
Load	$ \Gamma $	θ_r
 $Z_L = (r + jx)Z_0$	$\left[\frac{(r-1)^2 + x^2}{(r+1)^2 + x^2} \right]^{1/2}$	$\tan^{-1} \left(\frac{x}{r-1} \right) - \tan^{-1} \left(\frac{x}{r+1} \right)$
 Z_0	0 (no reflection)	irrelevant
 (short)	1	$\pm 180^\circ$ (phase opposition)
 (open)	1	0 (in-phase)
 $jX = j\omega L$	1	$\pm 180^\circ - 2 \tan^{-1} x$
 $jX = \frac{-j}{\omega C}$	1	$\pm 180^\circ - 2 \tan^{-1} x$

Figure 1-8: Magnitude and phase of the reflection coefficient for various types of loads.

Standing Waves

Using the relation $V_0^- = \Gamma V_0^+$ in Eqs. (1-46) and (1-47) yields

$$\tilde{V}(z) = V_0^+ (e^{-j\beta z} + \Gamma e^{j\beta z}), \quad (1-57)$$

$$\tilde{I}(z) = \frac{V_0^+}{Z_0} (e^{-j\beta z} - \Gamma e^{j\beta z}). \quad (1-58)$$

These expressions now contain only one (yet to be determined) unknown, V_0^+ . Before we proceed to solve for V_0^+ , however, let us examine the physical meaning underlying these expressions. We begin by deriving an expression for $|\tilde{V}(z)|$, which is the magnitude of $\tilde{V}(z)$. Upon using Eq. (1-56) in Eq. (1-57) and applying the relation $|\tilde{V}(z)| = [\tilde{V}(z)\tilde{V}^*(z)]^{1/2}$, where $\tilde{V}^*(z)$ is the complex conjugate of $\tilde{V}(z)$, we have

$$|\tilde{V}(z)| = \{ [V_0^+ (e^{-j\beta z} + |\Gamma|e^{j\theta_r}e^{j\beta z})] \quad (1-59)$$

$$\cdot [(V_0^+)^* (e^{j\beta z} + |\Gamma|e^{-j\theta_r}e^{-j\beta z})] \}^{1/2}$$

$$= |V_0^+| [1 + |\Gamma|^2 + |\Gamma|(e^{j(2\beta z + \theta_r)} + e^{-j(2\beta z + \theta_r)})]^{1/2}$$

$$= |V_0^+| [1 + |\Gamma|^2 + 2|\Gamma|\cos(2\beta z + \theta_r)]^{1/2}, \quad (1-60)$$

where we have used the identity

$$e^{jx} + e^{-jx} = 2 \cos x \quad (1-61)$$

for any real quantity x . To express the magnitude of $\tilde{V}$ as a function of distance d instead of z , we replace z with $-d$ on the right-hand side of Eq. (1-60):

$$|\tilde{V}(d)| = |V_0^+| [1 + |\Gamma|^2 + 2|\Gamma| \cos(2\beta d - \theta_r)]^{1/2}. \quad (1-62)$$

By applying the same steps to Eq. (1-58), a similar expression can be derived for $|\tilde{I}(d)|$, which is the magnitude of the current $I(d)$:

$$|\tilde{I}(d)| = \frac{|V_0^+|}{Z_0} [1 + |\Gamma|^2 - 2|\Gamma| \cos(2\beta d - \theta_r)]^{1/2}. \quad (1-63)$$

The variations of $|\tilde{V}(d)|$ and $|\tilde{I}(d)|$ as a function of d , which is the position on the line relative to the load (at $d = 0$), are illustrated in Fig. 1.9 for a line with $|V_0^+| = 1$ V, $|\Gamma| = 0.3$, $\theta_r = 30^\circ$, and $Z_0 = 50 \Omega$. The sinusoidal patterns are called *standing waves* and are caused by the interference of the two traveling waves. The maximum value of the standing-wave pattern of $\tilde{V}(d)$ corresponds to the position on the line at which the incident and reflected waves are *in-phase* [$2\beta d - \theta_r = 2n\pi$ in Eq. (1-62)]. Therefore, they add constructively to give a value equal to $(1 + |\Gamma|)|V_0^+| = 1.3$ V. The minimum value of $|\tilde{V}(d)|$ occurs when the two waves interfere destructively, which occurs when the incident and reflected waves are in *phase-opposition* [$2\beta d - \theta_r = (2n + 1)\pi$]. In this case, $|\tilde{V}(d)| = (1 - |\Gamma|)|V_0^+| = 0.7$ V.

► Whereas the repetition period is λ for the incident and reflected waves considered individually, the repetition period of the standing-wave pattern is $\lambda/2$.

The standing-wave pattern describes the spatial variation of the magnitude of $\tilde{V}(d)$ as a function of d . If one were to observe the variation of the instantaneous voltage as a function of time at location $d = d_{\max}$ in Fig. 1-9, the variation would be as $\cos \omega t$ and would have an amplitude equal to 1.3 V [i.e., $v(t)$ would oscillate between -1.3 V and $+1.3$ V]. Similarly, the instantaneous voltage $v(d, t)$ at any location d will be sinusoidal with amplitude equal to $|\tilde{V}(d)|$ at that d .

Close inspection of the voltage and current standing-wave patterns shown in Fig. 1.9 reveals that the two patterns are in phase opposition (when one is at a maximum, the other is at a minimum, and vice versa). This is a consequence of the fact that the third term in Eq. (1-62) is preceded by a plus sign, whereas the third term in Eq. (1-63) is preceded by a minus sign.

The standing-wave patterns shown in Fig. 1.9 are for a line with $\Gamma = 0.3 \angle 30^\circ$. The peak-to-peak variation of the pattern ($|\tilde{V}|_{\min} = (1 - |\Gamma|)|V_0^+|$ to $|\tilde{V}|_{\max} = (1 + |\Gamma|)|V_0^+|$) depends on $|\Gamma|$. For the special case of a matched line with $Z_L = Z_0$, we have $|\Gamma| = 0$ and $\tilde{V}(d) = V_0^+$ for all values of d , as shown in Fig. 1.10(a).

► With no reflected wave present, there is no interference and no standing waves.

The other end of the $|\Gamma|$ scale, at $|\Gamma| = 1$, corresponds to when the load is a short circuit ($\Gamma = -1$) or an open circuit ($\Gamma = 1$). The standing-wave patterns for those two cases are shown in Figs. 1.10(b) and (c); both exhibit maxima of $2|V_0^+|$ and minima equal

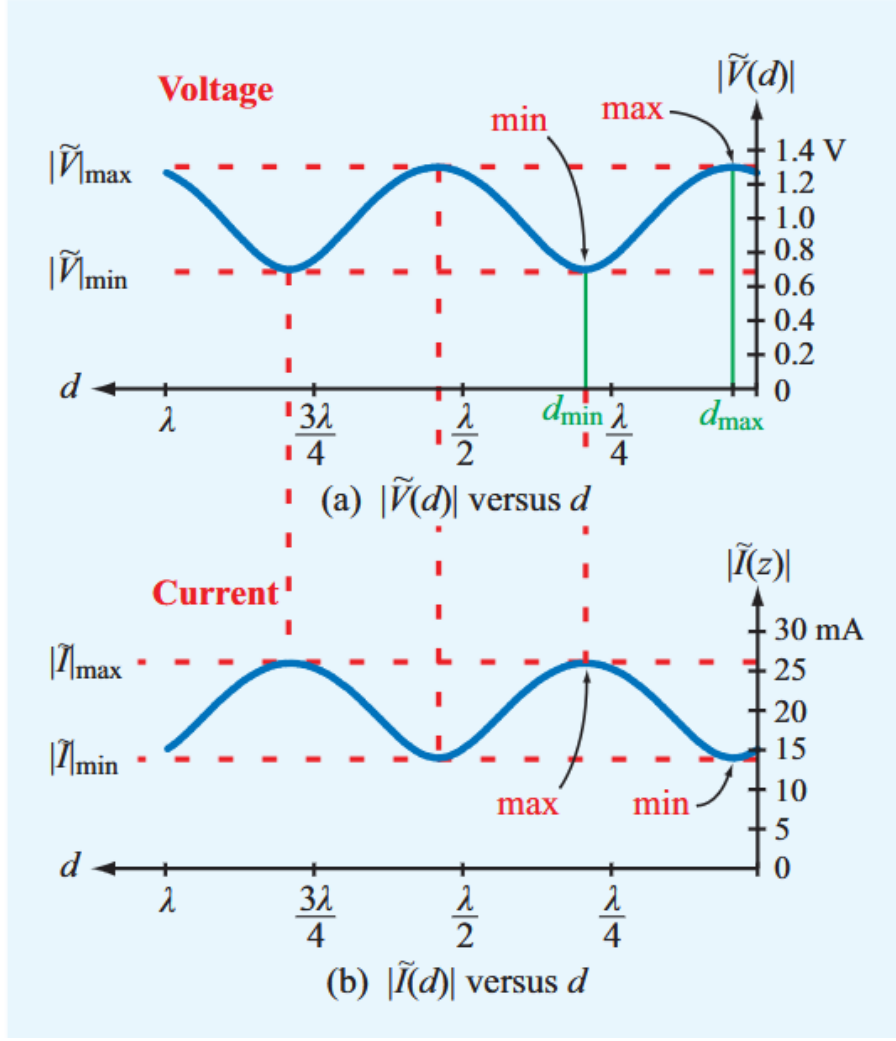


Figure 1-9: Standing-wave pattern for (a) $|\tilde{V}(d)|$ and (b) $|\tilde{I}(d)|$ for a lossless transmission line of characteristic impedance $Z_0 = 50 \, \Omega$, terminated in a load with a reflection coefficient $\Gamma = 0.3e^{j30^\circ}$. The magnitude of the incident wave $|V_0^+| = 1 \, \text{V}$. The standing-wave ratio is $S = |\tilde{V}|_{\max}/|\tilde{V}|_{\min} = 1.3/0.7 = 1.86$.

to zero, but the two patterns are spatially shifted relative to each other by a distance of $\lambda/4$. A purely reactive load (capacitor or inductor) also satisfies the condition $|\Gamma| = 1$, but θ_r is generally neither zero nor 180° (Fig 1.8).

Now let us examine the maximum and minimum values of the voltage magnitude. From Eq. (1-62), $|\tilde{V}(d)|$ is a maximum when the argument of the cosine function is equal to zero or a multiple of 2π . Let us denote $d_{\max}$ as the *distance from the load at which $|\tilde{V}(d)|$ is a maximum*. It then follows that

$$|\tilde{V}(d)| = |\tilde{V}|_{\max} = |V_0^+|(1 + |\Gamma|), \quad (1-64)$$

when

$$2\beta d_{\max} - \theta_r = 2n\pi, \quad (1-65)$$

with $n = 0$ or a positive integer. Solving Eq. (1-65) for $d_{\max}$, we have

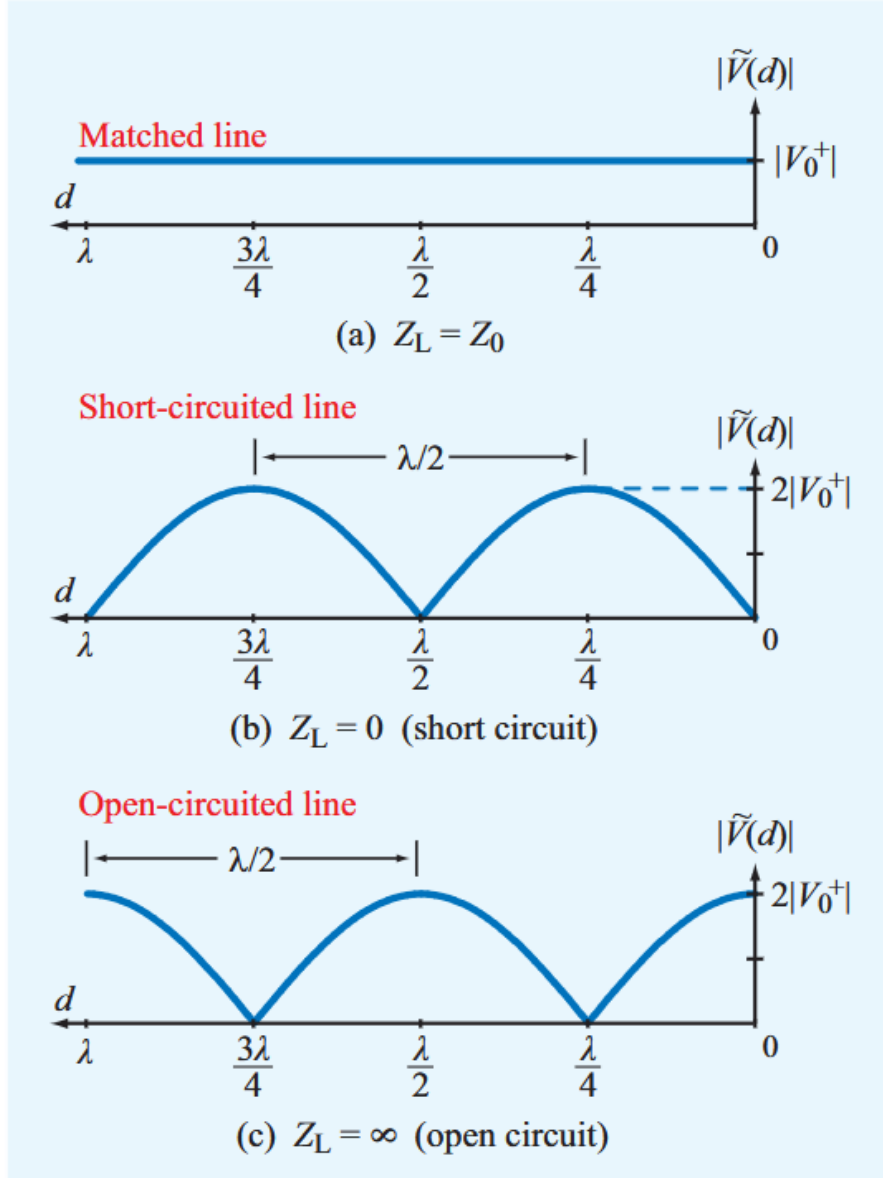


Figure 1-10: Voltage standing-wave patterns for (a) a matched load, (b) a short-circuited line, and (c) an open-circuited line

$$d_{\max} = \frac{\theta_r + 2n\pi}{2\beta} = \frac{\theta_r \lambda}{4\pi} + \frac{n\lambda}{2}, \quad (1-66)$$

where we have used $\beta = 2\pi/\lambda$. The phase angle of the voltage reflection coefficient, θ_r , is bounded between $-\pi$ and π radians. If $\theta_r \geq 0$, the first voltage maximum occurs at $d_{\max} = \theta_r \lambda / 4\pi$, corresponding to $n = 0$, but if $\theta_r < 0$, the first physically meaningful maximum occurs at $d_{\max} = (\theta_r \lambda / 4\pi) + \lambda/2$, corresponding to $n = 1$. Negative values of $d_{\max}$ correspond to locations past the end of the line; therefore, they have no physical significance.

Similarly, the minima of $|\tilde{V}(d)|$ occur at distances $d_{\min}$ where the argument of the cosine function in Eq. (1-62) is equal to $(2n + 1)\pi$, which gives

$$|\tilde{V}|_{\min} = |V_0^+|(1 - |\Gamma|), \quad (1-67)$$

when

$$(2\beta d_{\min} - \theta_r) = (2n + 1)\pi,$$

with $-\pi \leq \theta_r \leq \pi$. The first minimum corresponds to $n = 0$. The spacing between a maximum $d_{\max}$ and the adjacent minimum $d_{\min}$ is $\lambda/4$. Hence, the first minimum occurs at

$$d_{\min} = \begin{cases} d_{\max} + \lambda/4, & \text{if } d_{\max} < \lambda/4, \\ d_{\max} - \lambda/4, & \text{if } d_{\max} \geq \lambda/4. \end{cases} \quad (1-68)$$

► The locations on the line corresponding to voltage maxima correspond to current minima, and vice versa.

The ratio of $|\tilde{V}|_{\max}$ to $|\tilde{V}|_{\min}$ is called the *voltage standing-wave ratio* S , which from Eqs. (1-64) and (1-67) is given by

$$S = \frac{|\tilde{V}|_{\max}}{|\tilde{V}|_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|} \quad (\text{dimensionless}). \quad (1-69)$$

This quantity is often referred to by its acronym, VSWR or the shorter acronym SWR, and it provides a measure of the mismatch between the load and the transmission line:

► For a matched load with $\Gamma = 0$, we get $S = 1$, and for a line with $|\Gamma| = 1$, $S = \infty$.

Wave Impedance of the Lossless Line

The standing-wave patterns indicate that on a mismatched line the voltage and current magnitudes are oscillatory with position along the line and in phase opposition with each other. Hence, the voltage-to-current ratio, called the *wave impedance* $Z(d)$, must vary with position also. Using Eqs. (1-57) and (1-58) with $z = -d$,

$$\begin{aligned} Z(d) &= \frac{\tilde{V}(d)}{\tilde{I}(d)} = \frac{V_0^+ [e^{j\beta d} + \Gamma e^{-j\beta d}]}{\frac{V_0^+}{Z_0} [e^{j\beta d} - \Gamma e^{-j\beta d}]} \\ &= Z_0 \left[\frac{1 + \Gamma e^{-j2\beta d}}{1 - \Gamma e^{-j2\beta d}} \right] = Z_0 \left[\frac{1 + \Gamma_d}{1 - \Gamma_d} \right] \quad (\Omega), \end{aligned} \quad (1-70)$$

where we define

$$\Gamma_d = \Gamma e^{-j2\beta d} = |\Gamma| e^{j\theta_r} e^{-j2\beta d} = |\Gamma| e^{j(\theta_r - 2\beta d)} \quad (1-71)$$

as the *phase-shifted voltage reflection coefficient*, meaning that Γ_d has the same magnitude as Γ , but its phase is shifted by $2\beta d$ relative to that of Γ .

► $Z(d)$ is the ratio of the total voltage (incident- and reflected-wave voltages) to the total current at any point d on the line, in contrast with the characteristic impedance of the line Z_0 , which relates the voltage and current of each of the two waves individually ($Z_0 = V_0^+ / I_0^+ = -V_0^- / I_0^-$).

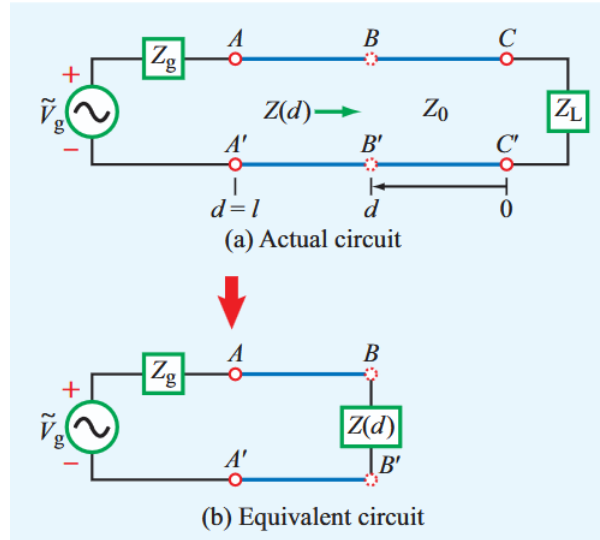


Figure 1-11: The segment to the right of terminals BB' can be replaced with a discrete impedance equal to the wave impedance $Z(d)$.

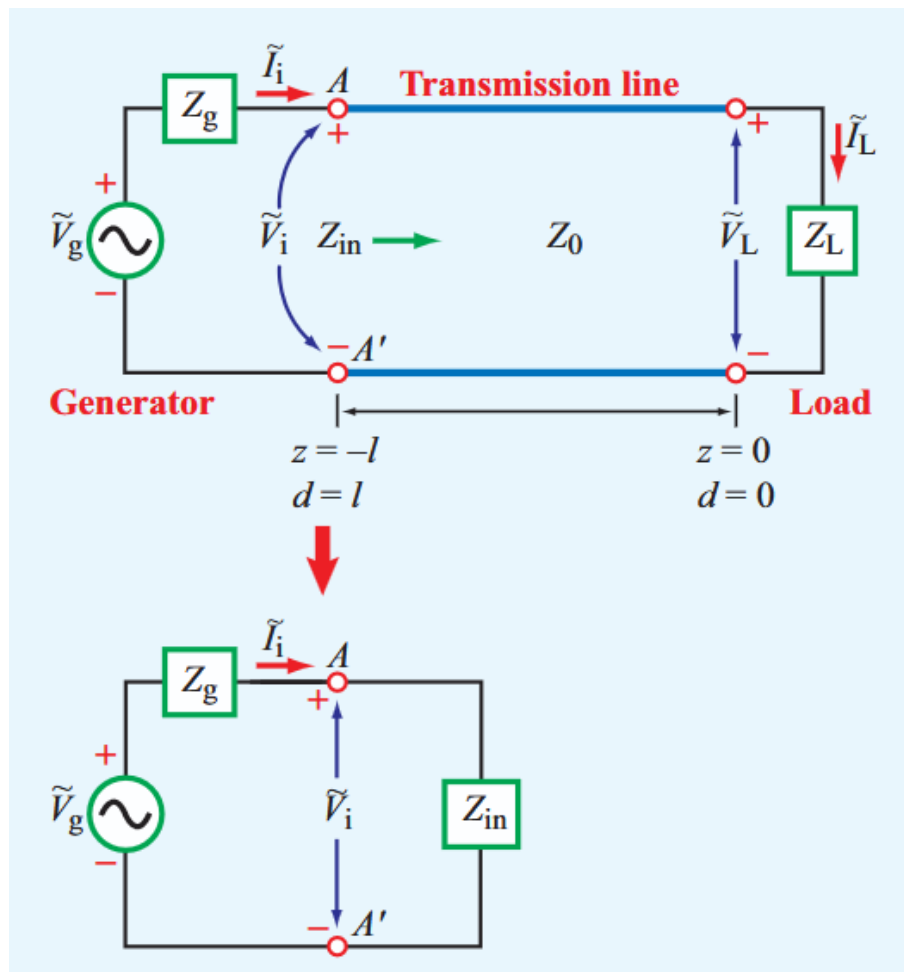


Figure 1-12: At the generator end the terminated transmission line can be replaced with the input impedance of the line Z_{in} .

In the circuit of Fig. 1.11(a) at terminals BB' at an arbitrary location d on the line, $Z(d)$ is the wave impedance of the line when “looking” to the right, i.e., towards the load. Application of the equivalence principle allows us to replace the segment to the right of terminals BB' with a lumped impedance of value $Z(d)$, as depicted in Fig. 1.11(b). From the standpoint of the input circuit to the left of terminals BB' , the two circuit configurations are electrically identical.

Of particular interest in many transmission-line problems is the *input impedance* at the source end of the line where $d = l$, which is given by

$$Z_{\text{in}} = Z(d = l) = Z_0 \left[\frac{1 + \Gamma_l}{1 - \Gamma_l} \right] \quad (1-72)$$

with

$$\Gamma_l = \Gamma e^{-j2\beta l} = |\Gamma| e^{j(\theta_r - 2\beta l)}. \quad (1-73)$$

By replacing Γ with Eq. (1.53) and using the relations

$$e^{j\beta l} = \cos \beta l + j \sin \beta l, \quad (1-74)$$

$$e^{-j\beta l} = \cos \beta l - j \sin \beta l, \quad (1-75)$$

Eq. (1-72) can be written in terms of the normalized load impedance z_L as

$$\begin{aligned} Z_{\text{in}} &= Z_0 \left(\frac{z_L \cos \beta l + j \sin \beta l}{\cos \beta l + j z_L \sin \beta l} \right) \\ &= Z_0 \left(\frac{z_L + j \tan \beta l}{1 + j z_L \tan \beta l} \right). \end{aligned} \quad (1-76)$$

From the standpoint of the generator circuit, the transmission line can be replaced with an impedance Z_{in} , as shown in Fig. 1.12. The phasor voltage across Z_{in} is given by

$$\tilde{V}_i = \tilde{I}_i Z_{\text{in}} = \frac{\tilde{V}_g Z_{\text{in}}}{Z_g + Z_{\text{in}}}. \quad (1-77)$$

Simultaneously, from the standpoint of the transmission line, the voltage across it at the input of the line is given by Eq. (1.57) with $z = -l$:

$$\tilde{V}_i = \tilde{V}(-l) = V_0^+ [e^{j\beta l} + \Gamma e^{-j\beta l}]. \quad (1-78)$$

Equating Eq. (1-77) to Eq. (1-78) and then solving for V_0^+ leads to

$$V_0^+ = \left(\frac{\tilde{V}_g Z_{\text{in}}}{Z_g + Z_{\text{in}}} \right) \left(\frac{1}{e^{j\beta l} + \Gamma e^{-j\beta l}} \right). \quad (1-79)$$

This completes the solution of the transmission-line wave equations given by Eqs. (1-17) and (1-19) for the special case of a lossless transmission line. We started out with the general solutions given by Eq. (1-23 and 1-24), which included four unknown amplitudes, V_0^+ , V_0^- , I_0^+ , and I_0^- . We then determined that $Z_0 = V_0^+/I_0^+ = -V_0^-/I_0^-$, thereby reducing the unknowns to the two voltage amplitudes only. Upon applying the boundary condition at the load, we were able to relate V_0^- to V_0^+ through Γ , and finally, by applying the boundary condition at the source, we obtained an expression for V_0^+ .

Special Cases of the Lossless Line

We often encounter situations involving lossless transmission lines with particular terminations or lines whose lengths lead to particularly useful line properties. We now consider some of these special cases.

Short-Circuited Line

The transmission line shown in Fig. 1.13(a) is terminated in a short circuit, $Z_L = 0$. Consequently, the voltage reflection coefficient defined by Eq. (1-53) is $\Gamma = -1$, and the voltage standing-wave ratio given by Eq. (1-69) is $S = \infty$. With $z = -d$ and $\Gamma = -1$ in Eqs. (1-57) and (1-58), and $\Gamma = -1$ in Eq. (1-70), the voltage, current, and wave impedance on a short-circuited lossless transmission line are given by

$$\tilde{V}_{sc}(d) = V_0^+ [e^{j\beta d} - e^{-j\beta d}] = 2jV_0^+ \sin \beta d, \quad (1-80)$$

$$\tilde{I}_{sc}(d) = \frac{V_0^+}{Z_0} [e^{j\beta d} + e^{-j\beta d}] = \frac{2V_0^+}{Z_0} \cos \beta d, \quad (1-81)$$

$$Z_{in}^{sc}(d) = \frac{\tilde{V}_{sc}(d)}{\tilde{I}_{sc}(d)} = jZ_0 \tan \beta d. \quad (1-82)$$

The voltage $\tilde{V}_{sc}(d)$ is zero at the load ($d = 0$), as it should be for a short circuit, and its amplitude varies as $\sin \beta d$. In contrast, the current $\tilde{I}_{sc}(d)$ is a maximum at the load, and it varies as $\cos \beta d$.

Both quantities are displayed in Fig. 1.13 as a function of d .

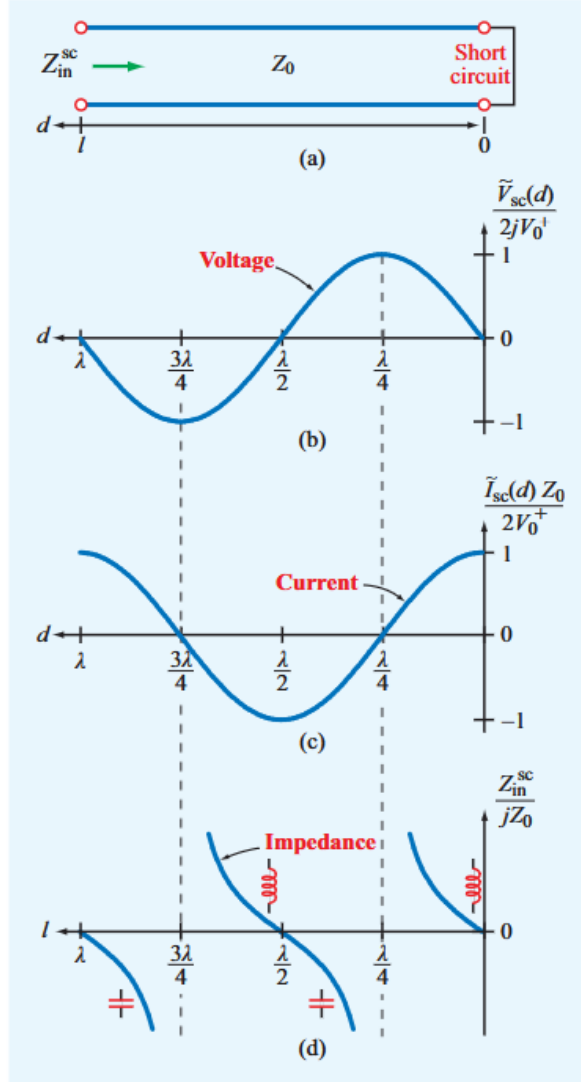


Figure 1-13: Transmission line terminated in a short circuit: (a) schematic representation, (b) normalized voltage on the line, (c) normalized current, and (d) normalized input impedance.

Denoting Z_{in}^{sc} as the *input impedance of a short-circuited line* of length l ,

$$Z_{in}^{sc} = \frac{\tilde{V}_{sc}(l)}{\tilde{I}_{sc}(l)} = jZ_0 \tan \beta l. \quad (1-83)$$

A plot of Z_{in}^{sc} / jZ_0 versus l is shown in Fig. 1.13(d). For the short-circuited line, if its length is less than $\lambda/4$, its impedance is equivalent to that of an inductor, and if it is between $\lambda/4$ and $\lambda/2$, it is equivalent to that of a capacitor.

In general, the input impedance Z_{in} of a line terminated in an arbitrary load has a real part, called the *input resistance* R_{in} , and an imaginary part, called the *input reactance* X_{in} :

$$Z_{in} = R_{in} + jX_{in}. \quad (1-84)$$

In the case of the short-circuited lossless line, the input impedance is purely reactive ($R_{in} = 0$). If $\tan \beta l \geq 0$, the line appears inductive to the source, acting like an equivalent

inductor L_{eq} whose impedance equals Z_{in}^{sc} . Thus,

$$j\omega L_{eq} = jZ_0 \tan \beta l, \quad \text{if } \tan \beta l \geq 0, \quad (1-85)$$

or

$$L_{eq} = \frac{Z_0 \tan \beta l}{\omega} \quad (\text{H}). \quad (1-86)$$

The minimum line length l that would result in an input impedance Z_{in}^{sc} equivalent to that of an inductor with inductance L_{eq} is

$$l = \frac{1}{\beta} \tan^{-1} \left(\frac{\omega L_{eq}}{Z_0} \right) \quad (\text{m}). \quad (1-87)$$

Similarly, if $\tan \beta l < 0$, the input impedance is capacitive, in which case the line acts like an equivalent capacitor with capacitance C_{eq} such that

$$\frac{1}{j\omega C_{eq}} = jZ_0 \tan \beta l, \quad \text{if } \tan \beta l < 0, \quad (1-88)$$

or

$$C_{eq} = \frac{1}{Z_0 \omega \tan \beta l} \quad (\text{F}). \quad (1-89)$$

Since l is a positive number, the shortest length l for which $\tan \beta l < 0$ corresponds to the range $\pi/2 \leq \beta l \leq \pi$. Hence, the minimum line length l that would result in an input impedance Z_{in}^{sc} equivalent to that of a capacitor of capacitance C_{eq} is

$$l = \frac{1}{\beta} \left[\pi - \tan^{-1} \left(\frac{1}{\omega C_{eq} Z_0} \right) \right] \quad (\text{m}). \quad (1-90)$$

► These results imply that, through proper choice of the length of a short-circuited line, we can make them into equivalent capacitors and inductors of any desired reactance.

Such a practice is indeed common in the design of microwave circuits and high-speed integrated circuits, because making an actual capacitor or inductor often is much more difficult than fabricating a microstrip transmission line with a shorted end on a circuit board.

Open-Circuited Line

With $Z_L = \infty$, as illustrated in Fig. 1.14(a), we have $\Gamma = 1$ and $S = \infty$, and the voltage, current, and input impedance are given by

$$\tilde{V}_{oc}(d) = V_0^+ [e^{j\beta d} + e^{-j\beta d}] = 2V_0^+ \cos \beta d, \quad (1-91)$$

$$\tilde{I}_{oc}(d) = \frac{V_0^+}{Z_0} [e^{j\beta d} - e^{-j\beta d}] = \frac{2jV_0^+}{Z_0} \sin \beta d, \quad (1-92)$$

$$Z_{in}^{oc} = \frac{\tilde{V}_{oc}(l)}{\tilde{I}_{oc}(l)} = -jZ_0 \cot \beta l. \quad (1-93)$$

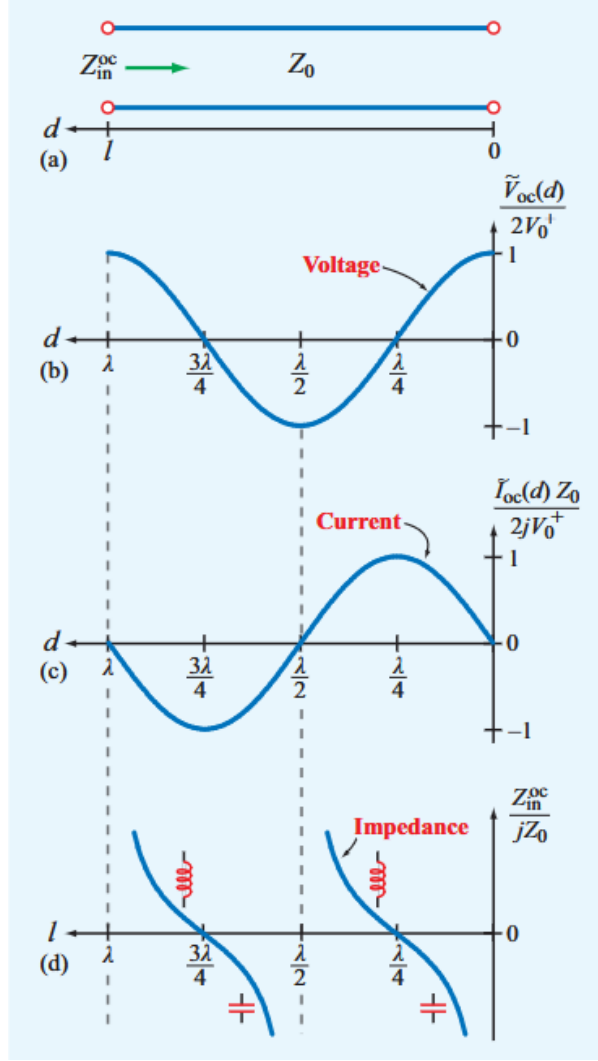


Figure 1-14: Transmission line terminated in an open circuit: (a) schematic representation, (b) normalized voltage on the line, (c) normalized current, and (d) normalized input impedance

Plots of these quantities are displayed in Fig. 1.14 as a function of d for the voltage and current and as a function of l for the input impedance.

Application of Short-Circuit/Open-Circuit Technique

A network analyzer is a radio-frequency (RF) instrument capable of measuring the impedance of any load connected to its input terminal. When used to measure (1) Z_{in}^{sc} , which is the input impedance of a lossless line terminated in a short circuit, and (2) Z_{in}^{oc} , which is the input impedance of the line when terminated in an open circuit, the combination of the two measurements can be used to determine the characteristic impedance of the line Z_0 and its phase constant β . Indeed, the product of the expressions given by Eqs. (1-83) and (1-93) gives

$$Z_0 = \pm \sqrt{Z_{in}^{sc} Z_{in}^{oc}}, \quad (1-94)$$

and the ratio of the same expressions leads to

$$\tan \beta l = \sqrt{-\frac{Z_{in}^{sc}}{Z_{in}^{oc}}}. \quad (1-95)$$

Because of the π phase ambiguity associated with the tangent function, the length l should be less than or equal to $\lambda/2$ to provide an unambiguous result.

Lines of Length $l = n\lambda/2$

If $l = n\lambda/2$, where n is an integer,

$$\tan \beta l = \tan \left[\left(\frac{2\pi}{\lambda} \right) \left(\frac{n\lambda}{2} \right) \right] = \tan n\pi = 0. \quad (1-96)$$

Consequently, Eq. (1-76) reduces to

$$Z_{in} = Z_L, \quad \text{for } l = n\lambda/2, \quad (1-97)$$

which means that a half-wavelength line (or any integer multiple of $\lambda/2$) does not modify the load impedance.

Quarter-Wavelength Transformer

Another case of interest is when the length of the line is a quarter-wavelength (or $\lambda/4 + n\lambda/2$, where $n = 0$ or a positive integer), corresponding to

$$\beta l = \left(\frac{2\pi}{\lambda} \right) \left(\frac{\lambda}{4} \right) = \frac{\pi}{2}.$$

From Eq. (1-76), the input impedance becomes

$$Z_{in} = \frac{Z_0^2}{Z_L}, \quad \text{for } l = \lambda/4 + n\lambda/2. \quad (1-98)$$

Matched Transmission Line: $Z_L = Z_0$

For a matched lossless transmission line with $Z_L = Z_0$,

- (1) the input impedance $Z_{in} = Z_0$ for all locations d on the line,
- (2) $\Gamma = 0$, and
- (3) all of the incident power is delivered to the load, regardless of the line length l .

A summary of the properties of standing waves is given in Fig 1.15.

Voltage maximum	$ \tilde{V} _{\max} = V_0^+ [1 + \Gamma]$
Voltage minimum	$ \tilde{V} _{\min} = V_0^+ [1 - \Gamma]$
Positions of voltage maxima (also positions of current minima)	$d_{\max} = \frac{\theta_r \lambda}{4\pi} + \frac{n\lambda}{2}, \quad n = 0, 1, 2, \dots$
Position of first maximum (also position of first current minimum)	$d_{\max} = \begin{cases} \frac{\theta_r \lambda}{4\pi}, & \text{if } 0 \leq \theta_r \leq \pi \\ \frac{\theta_r \lambda}{4\pi} + \frac{\lambda}{2}, & \text{if } -\pi \leq \theta_r \leq 0 \end{cases}$
Positions of voltage minima (also positions of current maxima)	$d_{\min} = \frac{\theta_r \lambda}{4\pi} + \frac{(2n+1)\lambda}{4}, \quad n = 0, 1, 2, \dots$
Position of first minimum (also position of first current maximum)	$d_{\min} = \frac{\lambda}{4} \left(1 + \frac{\theta_r}{\pi} \right)$
Input impedance	$Z_{\text{in}} = Z_0 \left(\frac{z_L + j \tan \beta l}{1 + j z_L \tan \beta l} \right) = Z_0 \left(\frac{1 + \Gamma_l}{1 - \Gamma_l} \right)$
Positions at which Z_{in} is real	at voltage maxima and minima
Z_{in} at voltage maxima	$Z_{\text{in}} = Z_0 \left(\frac{1 + \Gamma }{1 - \Gamma } \right)$
Z_{in} at voltage minima	$Z_{\text{in}} = Z_0 \left(\frac{1 - \Gamma }{1 + \Gamma } \right)$
Z_{in} of short-circuited line	$Z_{\text{in}}^{\text{sc}} = j Z_0 \tan \beta l$
Z_{in} of open-circuited line	$Z_{\text{in}}^{\text{oc}} = -j Z_0 \cot \beta l$
Z_{in} of line of length $l = n\lambda/2$	$Z_{\text{in}} = Z_L, \quad n = 0, 1, 2, \dots$
Z_{in} of line of length $l = \lambda/4 + n\lambda/2$	$Z_{\text{in}} = Z_0^2 / Z_L, \quad n = 0, 1, 2, \dots$
Z_{in} of matched line	$Z_{\text{in}} = Z_0$
$ V_0^+ $ = amplitude of incident wave; $\Gamma = \Gamma e^{j\theta_r}$ with $-\pi < \theta_r < \pi$; θ_r in radians; $\Gamma_l = \Gamma e^{-j2\beta l}$.	

Figure 1-15: Properties of standing waves on a lossless transmission line.

Power Flow on a Lossless Transmission Line

Our discussion thus far has focused on the voltage and current attributes of waves propagating on a transmission line. Now we examine the flow of power carried by the incident and reflected waves. We begin by reintroducing Eqs. (1-57) and (1-58) with $z = -d$:

$$\tilde{V}(d) = V_0^+ (e^{j\beta d} + \Gamma e^{-j\beta d}), \quad (1-99)$$

$$\tilde{I}(d) = \frac{V_0^+}{Z_0} (e^{j\beta d} - \Gamma e^{-j\beta d}). \quad (1-100)$$

In these expressions, the first terms represent the incident-wave voltage and current, and the terms involving Γ represent the reflected-wave voltage and current. The time-

domain expressions for the voltage and current at location d from the load are obtained by transforming Eq. (1-99) to the time domain:

$$v(d, t) = \Re[\tilde{V}e^{j\omega t}] \quad (1-101)$$

$$\begin{aligned} &= \Re \left[|V_0^+| e^{j\phi^+} (e^{j\beta d} + |\Gamma| e^{j\theta_r} e^{-j\beta d}) e^{j\omega t} \right] \\ &= |V_0^+| \left[\cos(\omega t + \beta d + \phi^+) + |\Gamma| \cos(\omega t - \beta d + \phi^+ + \theta_r) \right], \\ i(d, t) &= \frac{|V_0^+|}{Z_0} \left[\cos(\omega t + \beta d + \phi^+) - |\Gamma| \cos(\omega t - \beta d + \phi^+ + \theta_r) \right], \end{aligned} \quad (1-102)$$

where we used the relations $V_0^+ = |V_0^+| e^{j\phi^+}$ and $\Gamma = |\Gamma| e^{j\theta_r}$ that were introduced earlier as Eqs. (1-29) and (1-56), respectively.

Instantaneous Power

The instantaneous power carried by the transmission line is equal to the product of $v(d, t)$ and $i(d, t)$:

$$P(d, t) = v(d, t) i(d, t) : \quad (1-103)$$

$$\begin{aligned} &= |V_0^+| \left[\cos(\omega t + \beta d + \phi^+) + |\Gamma| \cos(\omega t - \beta d + \phi^+ + \theta_r) \right] \\ &\quad \times \frac{|V_0^+|}{Z_0} \left[\cos(\omega t + \beta d + \phi^+) - |\Gamma| \cos(\omega t - \beta d + \phi^+ + \theta_r) \right] \\ &= \frac{|V_0^+|^2}{Z_0} \left[\cos^2(\omega t + \beta d + \phi^+) - |\Gamma|^2 \cos^2(\omega t - \beta d + \phi^+ + \theta_r) \right] \quad (\text{W}). \end{aligned} \quad (1-104)$$

If the signs preceding ωt and βd in the argument of the cosine term are both positive or both negative, then the cosine term represents a wave traveling in the negative d direction.

Since d points from the load to the generator, the first term in Eq. (1-104) represents the *instantaneous incident power* traveling towards the load. This is the power that would be delivered to the load in the absence of a wave reflection (when $\Gamma = 0$). Because the second term has minus sign and the argument of the cosine of the second term in Eq. (1-104), that term represents the *instantaneous reflected power* traveling in the $+d$ direction—away from the load. Accordingly, we label these terms as

$$P^i(d, t) = \frac{|V_0^+|^2}{Z_0} \cos^2(\omega t + \beta d + \phi^+) \quad (\text{W}), \quad (1-105)$$

$$P^r(d, t) = -|\Gamma|^2 \frac{|V_0^+|^2}{Z_0} \cos^2(\omega t - \beta d + \phi^+ + \theta_r) \quad (\text{W}). \quad (1-106)$$

Using the trigonometric identity

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x), \quad (1-107)$$

the expressions in Eq. (1-105) can be rewritten as

$$P^i(d, t) = \frac{|V_0^+|^2}{2Z_0} [1 + \cos(2\omega t + 2\beta d + 2\phi^+)] , \quad (1-108)$$

$$P^r(d, t) = -|\Gamma|^2 \frac{|V_0^+|^2}{2Z_0} [1 + \cos(2\omega t - 2\beta d + 2\phi^+ + 2\theta_r)] . \quad (1-109)$$

We note that in each case, the instantaneous power consists of a dc term (not varying in time) and an ac term that oscillates at an angular frequency of 2ω .

► The power oscillates at twice the rate of the voltage or current.

Time-Average Power

From a practical standpoint, we usually are more interested in the time-average power flowing along the transmission line, $P_{av}(d)$, than in the instantaneous power $P(d, t)$. To compute $P_{av}(d)$, we can use a time-domain approach or a computationally simpler phasor-domain approach. For completeness, we consider both.

Time-Domain Approach

The time-average power is equal to the instantaneous power averaged over one time period $T = 1/f = 2\pi/\omega$. For the incident wave, its time-average power is

$$P_{av}^i(d) = \frac{1}{T} \int_0^T P^i(d, t) dt = \frac{\omega}{2\pi} \int_0^{2\pi/\omega} P^i(d, t) dt. \quad (1-110)$$

Upon inserting Eq. (1-105) into Eq. (1-110) and performing the integration, we obtain

$$P_{av}^i = \frac{|V_0^+|^2}{2Z_0} \quad (\text{W}), \quad (1-111)$$

which is identical with the dc term of $P^i(d, t)$ given by Eq. (1-105). A similar treatment for the reflected wave gives

$$P_{av}^r = -|\Gamma|^2 \frac{|V_0^+|^2}{2Z_0} = -|\Gamma|^2 P_{av}^i. \quad (1-112)$$

► The average reflected power is equal to the average incident power, diminished by a multiplicative factor of $|\Gamma|^2$.

Note that the expressions for P_{av}^i and P_{av}^r are independent of d , which means that the time-average powers carried by the incident and reflected waves do not change as they travel along the transmission line. This is as expected, because the transmission line is lossless.

The *net average power flowing towards (and then absorbed by) the load* shown in Fig. 1.16 is

$$P_{av} = P_{av}^i + P_{av}^r = \frac{|V_0^+|^2}{2Z_0} [1 - |\Gamma|^2] \quad (\text{W}). \quad (1-113)$$

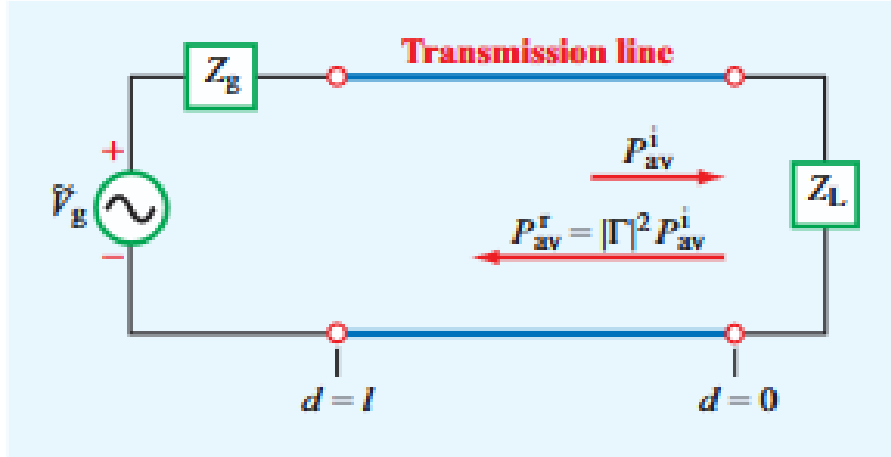


Figure 1-16: The time-average power reflected by a load connected to a lossless transmission line is equal to the incident power multiplied by $|\Gamma|^2$.

Phasor-Domain Approach

For any propagating wave with voltage and current phasors $\tilde{V}$ and $\tilde{I}$, a useful formula for computing the time-average power is

$$P_{av} = \frac{1}{2} \Re [\tilde{V} \cdot \tilde{I}^*], \quad (1-114)$$

where $\tilde{I}^*$ is the complex conjugate of $\tilde{I}$. Application of this formula to Eqs. (1-99) and (1-100) gives

$$\begin{aligned} P_{av} &= \frac{1}{2} \Re \left[V_0^+ (e^{j\beta d} + \Gamma e^{-j\beta d}) \frac{(V_0^+)^*}{Z_0} (e^{-j\beta d} - \Gamma^* e^{j\beta d}) \right] \\ &= \frac{1}{2} \Re \left[\frac{|V_0^+|^2}{Z_0} (1 - \Gamma^* e^{-j2\beta d} - \Gamma e^{j2\beta d}) \right] \\ &= \frac{|V_0^+|^2}{2Z_0} [(1 - |\Gamma|^2) + \Re\{|\Gamma| [e^{-j(2\beta d - \theta_r)} - |\Gamma| e^{j(2\beta d - \theta_r)}]\}] \\ &= \frac{|V_0^+|^2}{2Z_0} [1 - |\Gamma|^2 + |\Gamma| [\cos(2\beta d - \theta_r) - \cos(2\beta d - \theta_r)]] \\ &= \frac{|V_0^+|^2}{2Z_0} [1 - |\Gamma|^2], \end{aligned} \quad (1-115)$$

which is identical to Eq. (1-113).

The Smith Chart

The *Smith chart*, developed by P. H. Smith in 1939, is a widely used graphical tool for analyzing and designing transmission-line circuits. Even though it was originally intended to facilitate calculations involving complex impedances, the Smith chart has become an important avenue for comparing and characterizing the performance of microwave circuits. As the material in this and the next section demonstrates, use of the Smith

chart not only avoids tedious manipulations of complex numbers, but it also allows an engineer to design impedance-matching circuits with relative ease.

Parametric Equations

The reflection coefficient Γ is, in general, a complex quantity composed of a magnitude $|\Gamma|$ and a phase angle θ_r , or, equivalently, a real part Γ_r and an imaginary part Γ_i ,

$$\Gamma = |\Gamma|e^{j\theta_r} = \Gamma_r + j\Gamma_i, \quad (1-116)$$

where

$$\Gamma_r = |\Gamma| \cos \theta_r, \quad (1-117)$$

$$\Gamma_i = |\Gamma| \sin \theta_r. \quad (1-118)$$

The Smith chart lies in the complex Γ plane. In Fig. 1.17, point A represents a reflection coefficient $\Gamma_A = 0.3 + j0.4$ or, equivalently,

$$|\Gamma_A| = [(0.3)^2 + (0.4)^2]^{1/2} = 0.5$$

and

$$\theta_{rA} = \tan^{-1}(0.4/0.3) = 53^\circ.$$

Similarly, point B represents $\Gamma_B = -0.5 - j0.2$, or $|\Gamma_B| = 0.54$ and $\theta_{rB} = 202^\circ$ or, equivalently,

$$\theta_{rB} = (360^\circ - 202^\circ) = -158^\circ.$$

► When both Γ_r and Γ_i are negative, θ_r is in the third quadrant in the Γ_r - Γ_i plane. Thus, when using $\theta_r = \tan^{-1}(\Gamma_i/\Gamma_r)$ to compute θ_r , it may be necessary to add or subtract 180° to obtain the correct value of θ_r .

The *unit circle* shown in Fig. 1.17 corresponds to $|\Gamma| = 1$. Because $|\Gamma| \leq 1$ for a transmission line terminated with a passive load, only that part of the Γ_r - Γ_i plane that lies within the unit circle is useful to us; hence, future drawings will be limited to the domain contained within the unit circle.

Impedances on a Smith chart are represented by their values normalized to the line's characteristic impedance, Z_0 . From

$$\Gamma = \frac{Z_L/Z_0 - 1}{Z_L/Z_0 + 1} = \frac{z_L - 1}{z_L + 1}, \quad (1-119)$$

the inverse relation is

$$z_L = \frac{1 + \Gamma}{1 - \Gamma}. \quad (1-120)$$

The normalized load impedance z_L is, in general, a complex quantity composed of a *normalized load resistance* r_L and a *normalized load reactance* x_L :

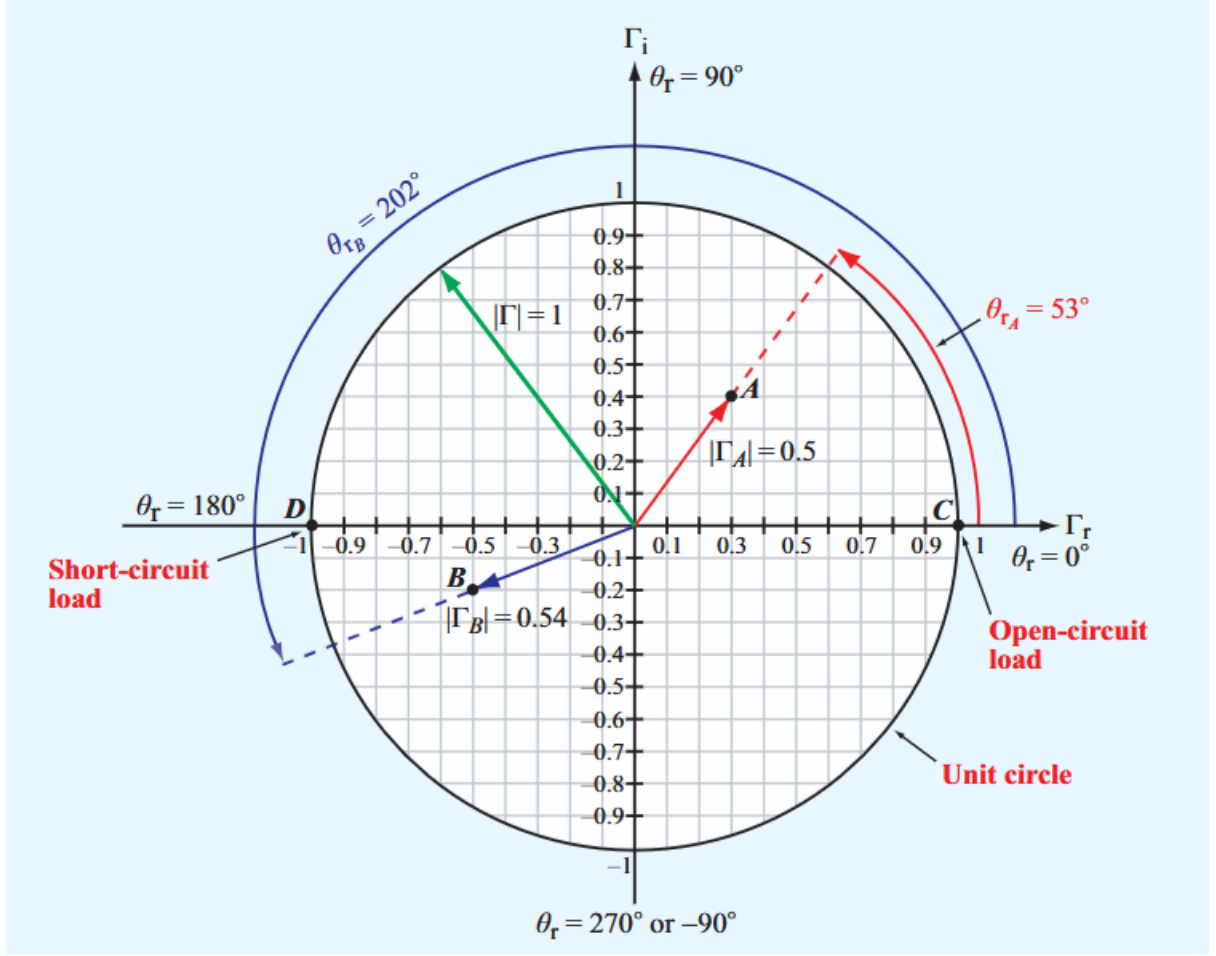


Figure 1-17: The complex Γ plane. Point A is at $\Gamma_A = 0.3 + j0.4 = 0.5e^{j53^\circ}$, and point B is at $\Gamma_B = -0.5 - j0.2 = 0.54e^{j202^\circ}$. The unit circle corresponds to $|\Gamma| = 1$. At point C , $\Gamma = 1$, corresponding to an open-circuit load, and at point D , $\Gamma = -1$, corresponding to a short circuit.

$$z_L = r_L + jx_L. \quad (1-121)$$

Using Eqs. (1-116) and (1-121) in Eq. (1-120), we have

$$r_L + jx_L = \frac{(1 + \Gamma_r) + j\Gamma_i}{(1 - \Gamma_r) - j\Gamma_i}, \quad (1-122)$$

which can be manipulated to obtain explicit expressions for each of r_L and x_L in terms of Γ_r and Γ_i . This is accomplished by multiplying the numerator and denominator of the right-hand side of Eq. (1-122) by the complex conjugate of the denominator and then separating the result into real and imaginary parts. These steps lead to

$$r_L = \frac{1 - \Gamma_r^2 - \Gamma_i^2}{(1 - \Gamma_r)^2 + \Gamma_i^2}, \quad (1-123)$$

$$x_L = \frac{2\Gamma_i}{(1 - \Gamma_r)^2 + \Gamma_i^2}. \quad (1-124)$$

Equation (1-123) implies that there exist many combinations of values for Γ_r and Γ_i that yield the same value for the normalized load resistance r_L . For example, $(\Gamma_r, \Gamma_i) = (0.33, 0)$ gives $r_L = 2$, as does $(\Gamma_r, \Gamma_i) = (0.5, 0.29)$, as well as an infinite number of other combinations. In fact, if we were to plot in the Γ_r - Γ_i plane all possible combinations of Γ_r and Γ_i corresponding to $r_L = 2$, we would obtain the circle labeled $r_L = 2$ in Fig. 1.18.

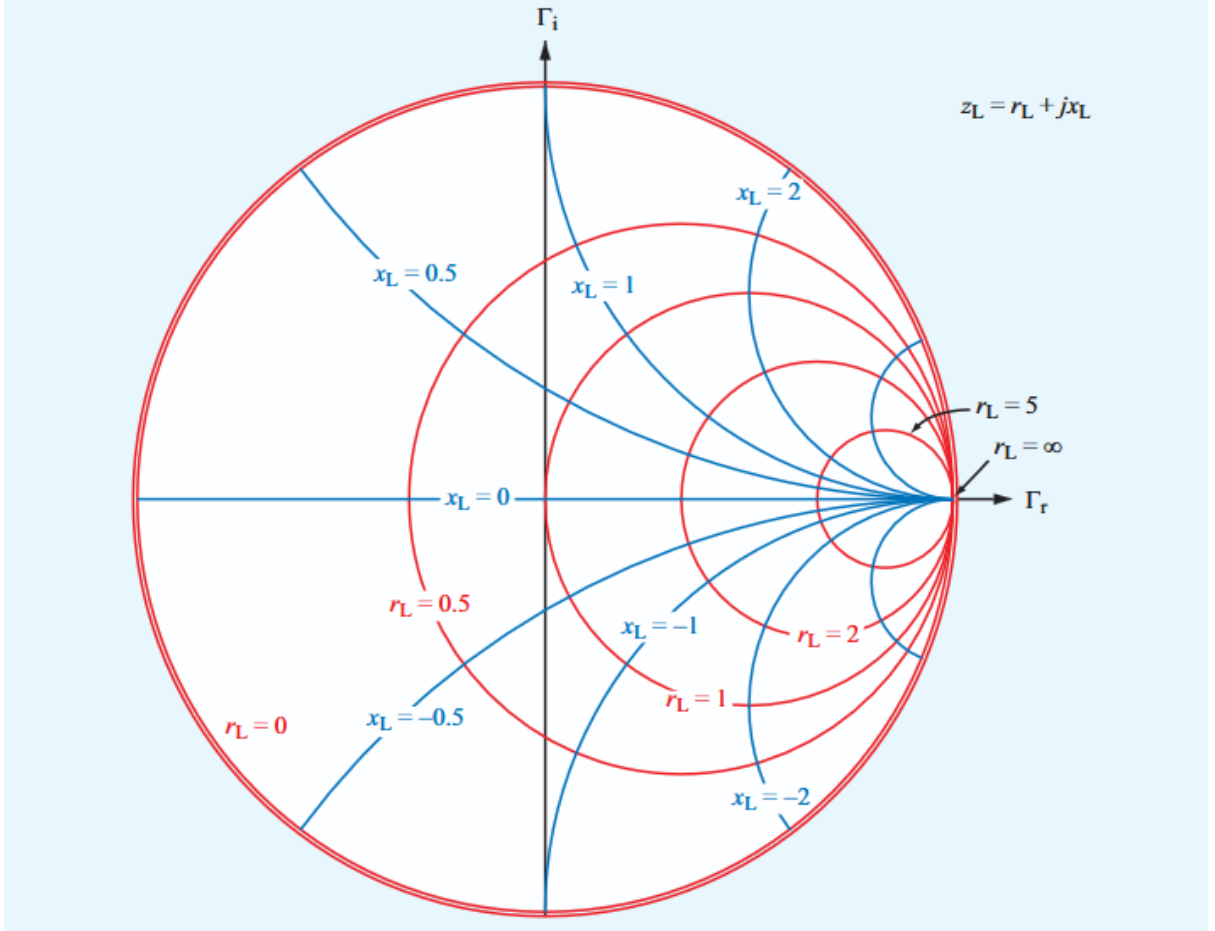


Figure 1-18: Families of r_L and x_L circles within the domain $|\Gamma| \leq 1$. The r_L circles are concentric with the outermost circle corresponding to $r_L = 0$ and the innermost circle (of zero radius) corresponding to $r_L = \infty$. The parts of the x_L circles contained in the Smith chart are in the upper half of the chart if x_L is positive and in the lower half of the chart if x_L is negative.

Similar circles can be obtained for other values of r_L . After some algebraic manipulations, Eq. (1-23) can be rearranged into the *parametric equation* for the circle in the Γ_r - Γ_i plane corresponding to a given value of r_L :

$$\left(\Gamma_r - \frac{r_L}{1 + r_L}\right)^2 + \Gamma_i^2 = \left(\frac{1}{1 + r_L}\right)^2. \quad (1-125)$$

The standard equation for a circle in the x - y plane with center at (x_0, y_0) and radius a is

$$(x - x_0)^2 + (y - y_0)^2 = a^2. \quad (1-126)$$

Comparison of Eq. (1-125) with Eq. (1-126) shows that the r_L circle is centered at $\Gamma_r = r_L/(1 + r_L)$ and $\Gamma_i = 0$, and its radius is $1/(1 + r_L)$. It therefore follows that all r_L circles pass through the point $(\Gamma_r, \Gamma_i) = (1, 0)$. The largest circle shown in Fig. 1.18 is for $r_L = 0$, which also is the unit circle corresponding to $|\Gamma| = 1$. This is to be expected, because when $r_L = 0$, $|\Gamma| = 1$ regardless of the magnitude of x_L .

A similar manipulation of the expression for x_L given by Eq. (1-124) leads to

$$(\Gamma_r - 1)^2 + \left(\Gamma_i - \frac{1}{x_L}\right)^2 = \left(\frac{1}{x_L}\right)^2, \quad (1-127)$$

which is the equation of a circle of radius $(1/x_L)$ centered at $(\Gamma_r, \Gamma_i) = (1, 1/x_L)$. The x_L circles in the Γ_r - Γ_i plane are quite different from those for constant r_L . To start with, the normalized reactance x_L may assume both positive and negative values, whereas the normalized resistance cannot be negative (negative resistances cannot be realized in passive circuits). Hence, Eq. (1-127) represents two families of circles: one for positive values of x_L and another for negative ones. Furthermore, as shown in Fig. 1.18, only part of a given circle falls within the bounds of the $|\Gamma| = 1$ unit circle.

The families of circles of the two parametric equations given by Eqs. (1-125) and (1-127) plotted for selected values of r_L and x_L constitute the Smith chart shown in Fig. 1.19. The Smith chart provides a graphical evaluation of Eqs. (1-123) and (1-124) and their inverses.

► Each point in the Smith chart conveys the values of two interrelated quantities, namely z_L and Γ . The intersection of the r_L and x_L circles at that point defines $z_L = r_L + jx_L$. Simultaneously, the location of the point defines $|\Gamma|$ and θ_r .

For example, point P in Fig. 1.19 represents a normalized load impedance $z_L = 2 - j1$, which corresponds to a voltage reflection coefficient $\Gamma = 0.45 \exp(-j26.6^\circ)$. The magnitude $|\Gamma| = 0.45$ is obtained by dividing the length of the line between the center of the Smith chart and point P (designated OP in Fig. 1.19), by OR , which is the length of the line between the center of the Smith chart and the edge of the unit circle (the radius of the unit circle corresponds to $|\Gamma| = 1$). The perimeter of the Smith chart contains three concentric scales. The innermost scale is labeled *angle of reflection coefficient in degrees*. This is the scale for θ_r . As indicated in Fig. 1.19, $\theta_r = -26.6^\circ$ (-0.46 rad) for point P . The meanings and uses of the other two scales are discussed next.

Impedance Matching

A transmission line usually connects a generator circuit at one end to a load at the other. The load may be an antenna, a computer terminal, or any circuit with an equivalent input impedance Z_L .

► The transmission line is said to be matched to the load when its characteristic impedance $Z_0 = Z_L$, in which case waves traveling on the line towards the load are not reflected back to the source.

Since the primary use of a transmission line is to transfer power or transmit coded signals (such as digital data), a matched load ensures that all of the power delivered to

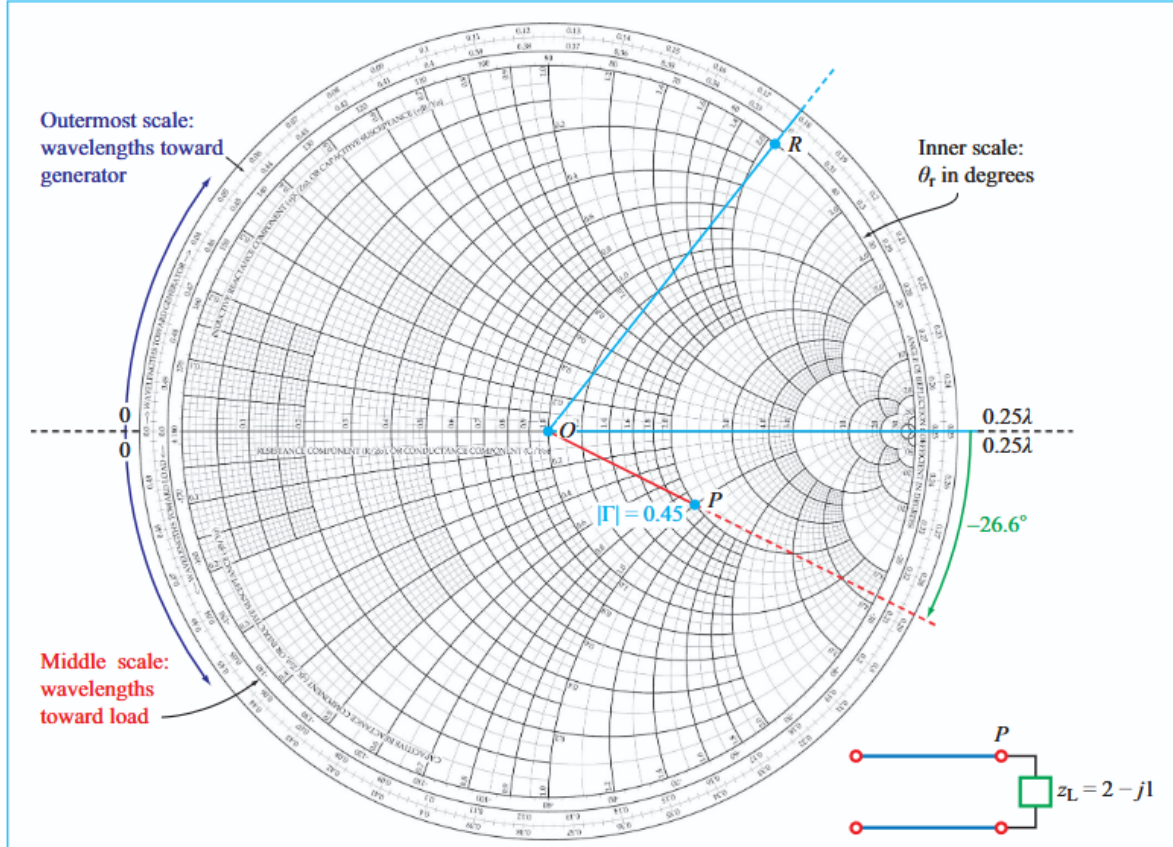


Figure 1-19: Point P represents a normalized load impedance $z_L = 2 - j1$. The reflection coefficient has a magnitude $|\Gamma| = OP/OR = 0.45$ and an angle $\theta_r = -26.6^\circ$. Point R is an arbitrary point on the $r_L = 0$ circle (which also is the $|\Gamma| = 1$ circle).

the transmission line by the source is transferred to the load (and no *echoes* are relayed back to the source).

The simplest solution to matching a load to a transmission line is to design the load circuit such that its impedance

$$Z_L = Z_0.$$

Unfortunately, this may not be possible in practice because the load circuit may have to satisfy other requirements. An alternative solution is to place an *impedance-matching network* between the load and the transmission line as illustrated in Fig. 1.20.

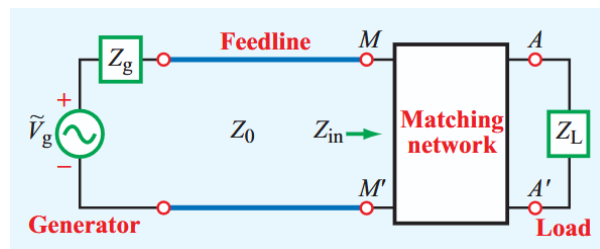


Figure 1-20: The function of a matching network is to transform the load impedance Z_L such that the input impedance Z_{in} looking into the network is equal to Z_0 of the feedline.

► The purpose of the matching network is to eliminate reflections at terminals MM' for waves incident from the source. Even though multiple reflections may occur between AA' and MM' , only a forward traveling wave exists on the feedline.

Within the matching network, reflections can occur at both terminals (AA' and MM'), creating a standing-wave pattern, but the net result (of all of the multiple reflections within the matching network) is that the wave incident from the source experiences no reflection when it reaches terminals MM' . This is achieved by designing the matching network to exhibit an impedance equal to Z_0 at MM' when looking into the network from the transmission line side. If the network is lossless, then all of the power going into it will end up in the load.

► Matching networks may consist of lumped elements, such as capacitors and inductors (but not resistors because resistors incur ohmic losses), or of sections of transmission lines with appropriate lengths and terminations.

The matching network, which is intended to match a load impedance

$$Z_L = R_L + jX_L$$

to a lossless transmission line with characteristic impedance Z_0 , may be inserted either in series (between the load and the feedline) as in Fig. 1.21(a) and (b) or in parallel [Fig. 1.21(c) to (e)]. In either case, the network has to transform the real part of the load impedance from R_L (at the load) to Z_0 at MM' in Fig. 1.20 and transform the reactive part from X_L (at the load) to zero at MM' . To achieve these two transformations, the matching network must have at least two degrees of freedom (that is, two adjustable parameters).

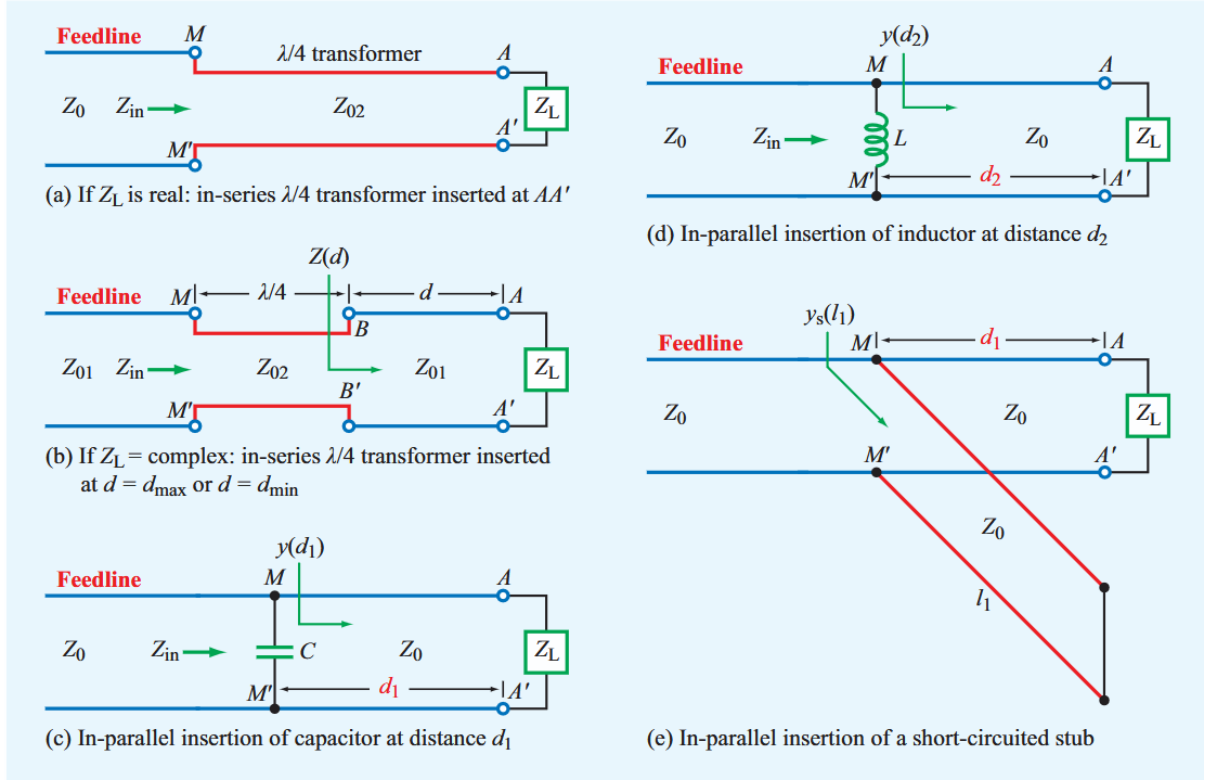


Figure 1-21: Five examples of in-series and in-parallel matching networks

If $X_L = 0$, the problem reduces to a single transformation, in which case matching can be realized by inserting a quarter-wavelength transformer next to the load [Fig. 1.21(a)].

► For the general case where $X_L \neq 0$, a $\lambda/4$ transformer can be designed to provide the desired matching, but it has to be inserted at a distance $d_{\max}$ or $d_{\min}$ from the load [Fig. 1.21(b)], where $d_{\max}$ and $d_{\min}$ are the distances to voltage maxima and minima, respectively.

Lumped-Element Matching

In the arrangement shown in Fig. 1.22, the matching network consists of a single lumped element, either a capacitor or an inductor, connected in parallel with the line at a distance d from the load. Parallel connections call for working in the admittance domain. Hence, the load is denoted by an admittance Y_L , and the line has a characteristic admittance Y_0 . The shunt element has an admittance Y_s . At MM' , Y_d is the admittance due to the transmission-line segment to the right of MM' . The input admittance Y_{in} (referenced at a point just to the left of MM') is equal to the sum of Y_d and Y_s :

$$Y_{in} = Y_d + Y_s. \quad (1-128)$$

In general, Y_d is complex, and Y_s is purely imaginary because it represents a reactive element (capacitor or inductor). Hence, Eq. (1-128) can be written as

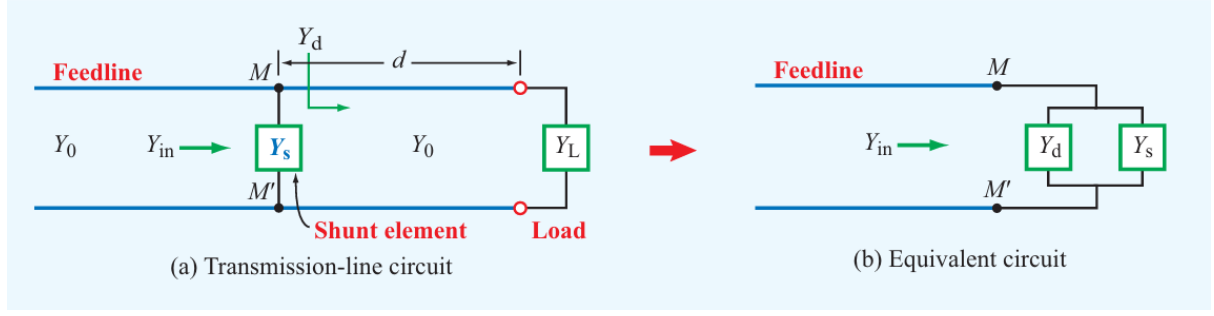


Figure 1-22: Inserting a reactive element with admittance Y_s at MM' modifies Y_d to Y_{in} .

$$Y_{in} = (G_d + jB_d) + jB_s = G_d + j(B_d + B_s). \quad (1-129)$$

When all quantities are normalized to Y_0 , Eq. (1-129) becomes

$$y_{in} = g_d + j(b_d + b_s). \quad (1-130)$$

To achieve a matched condition at MM' , it is necessary that

$$y_{in} = 1 + j0,$$

which translates into two specific conditions, namely

$$g_d = 1 \quad (\text{real-part condition}), \quad (1-131)$$

$$b_s = -b_d \quad (\text{imaginary-part condition}). \quad (1-132)$$

The real-part condition is realized through the choice of d , which is the distance from the load to the shunt element, and the imaginary-part condition is realized through the choice of the lumped element (capacitor or inductor) and its value. These two choices are the two degrees of freedom needed in order to match the load to the feedline.

Single-Stub Matching

The single-stub matching network shown in Fig. 1.23(a) consists of two transmission line sections: one of length d connecting the load to the feedline at MM' and another of length l connected in parallel with the other two lines at MM' . This second line is called a *stub*, and it is usually terminated in either a short or open circuit; hence, its input impedance and admittance are purely reactive. The stub shown in Fig. 1.23(a) has a short-circuit termination.

► The required two degrees of freedom are provided by the length l of the stub and the distance d from the load to the stub position.

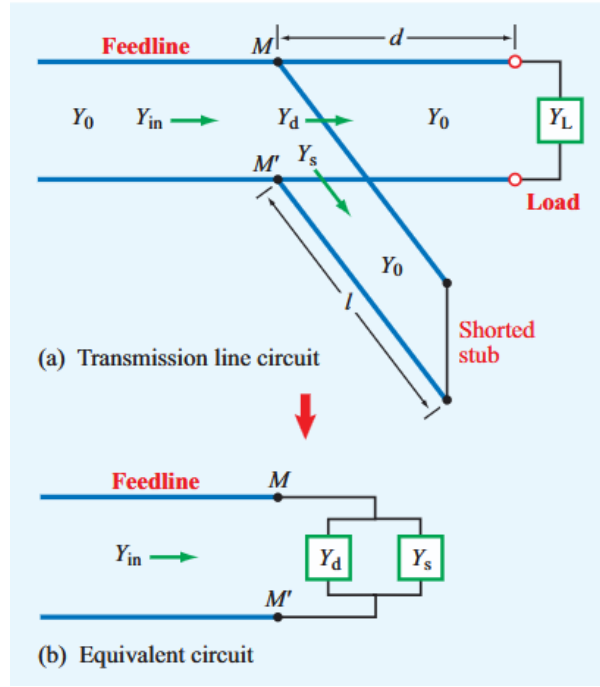


Figure 1-23: Shorted-stub matching network.

Because at MM' the stub is added in parallel to the line (which is why it is called a *shunt stub*), it is easier to work with admittances than with impedances. The matching procedure consists of two steps. In the first step, the distance d is selected to transform the load admittance,

$$Y_L = \frac{1}{Z_L},$$

into an admittance of the form

$$Y_d = Y_0 + jB$$

when looking toward the load at MM' . Then in the second step, the length l of the stub line is selected so that its input admittance Y_s at MM' is equal to $-jB$. The parallel sum of the two admittances at MM' yields Y_0 , which is the characteristic admittance of the line.

The Complete Smith Chart

Black Magic Design

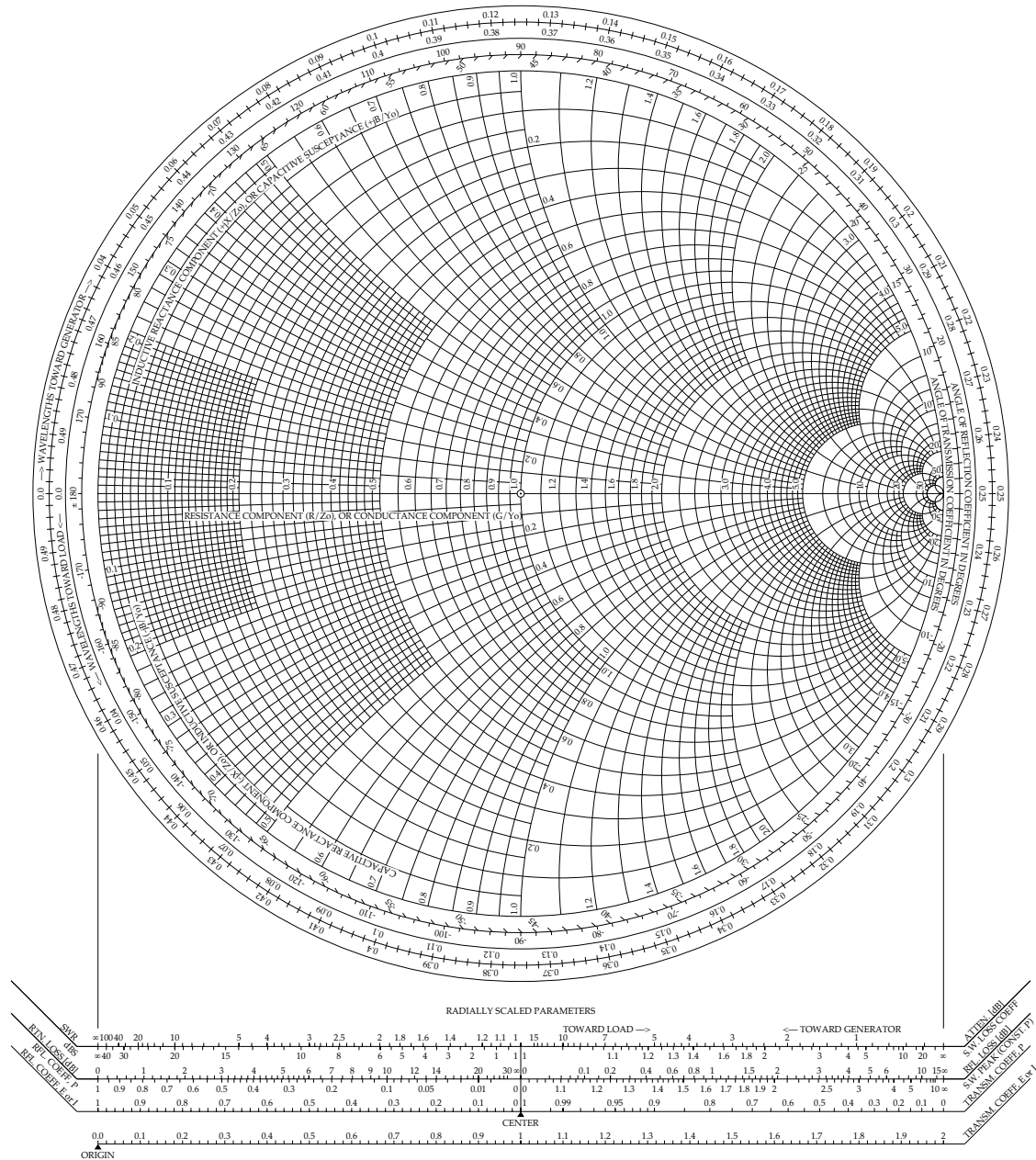


Figure 1-24: Sample Smith Chart

Example-1:

An **air line** is a transmission line in which air separates the two conductors, which renders $G' = 0$ because $\sigma = 0$. In addition, assume that the conductors are made of a material with high conductivity so that $R' \approx 0$. For an air line with a characteristic impedance of 50Ω and a phase constant of 20 rad/m at 700 MHz , find the line inductance L' and the line capacitance C' .

Solution:

The following quantities are given:

$$\begin{aligned} Z_0 &= 50 \Omega, \\ \beta &= 20 \text{ rad/m}, \\ f &= 700 \text{ MHz} = 7 \times 10^8 \text{ Hz}. \end{aligned}$$

With $R' = G' = 0$,

$$\begin{aligned} \beta &= \Im \left[\sqrt{(j\omega L')(j\omega C')} \right] = \Im \left(j\omega \sqrt{L'C'} \right) = \omega \sqrt{L'C'}, \\ Z_0 &= \sqrt{\frac{j\omega L'}{j\omega C'}} = \sqrt{\frac{L'}{C'}}. \end{aligned}$$

The ratio of β to Z_0 is $\beta/Z_0 = \omega C'$, or

$$\begin{aligned} C' &= \frac{\beta}{\omega Z_0} = \frac{20}{2\pi \times 7 \times 10^8 \times 50} = 9.09 \times 10^{-11} \text{ (F/m)} \\ &= 90.9 \text{ (pF/m)}. \end{aligned}$$

From $Z_0 = \sqrt{L'/C'}$, it follows that

$$\begin{aligned} L' &= Z_0^2 C' = (50)^2 \times 90.9 \times 10^{-12} = 2.27 \times 10^{-7} \text{ (H/m)} \\ &= 227 \text{ (nH/m)}. \end{aligned}$$

Example-2

A 100Ω transmission line is connected to a load consisting of a 50Ω resistor in series with a 10 pF capacitor. Find the reflection coefficient at the load for a 100 MHz signal.

Solution:

$$\begin{aligned} R_L &= 50 \Omega, \quad C_L = 10 \text{ pF} = 10^{-11} \text{ F}, \\ Z_0 &= 100 \Omega, \quad f = 100 \text{ MHz} = 10^8 \text{ Hz}. \end{aligned}$$

The normalized load impedance is

$$\begin{aligned} z_L &= \frac{Z_L}{Z_0} \\ &= \frac{R_L - j/(\omega C_L)}{Z_0} \\ &= \frac{1}{100} \left(50 - j \frac{1}{2\pi \times 10^8 \times 10^{-11}} \right) \\ &= (0.5 - j1.59) \Omega. \end{aligned}$$

The voltage reflection coefficient is

$$\begin{aligned}
 \Gamma &= \frac{z_L - 1}{z_L + 1} \\
 &= \frac{0.5 - j1.59 - 1}{0.5 - j1.59 + 1} \\
 &= \frac{-0.5 - j1.59}{1.5 - j1.59} \\
 &= \frac{-1.67e^{j72.6^\circ}}{2.19e^{-j46.7^\circ}} \\
 &= -0.76e^{j119.3^\circ}.
 \end{aligned}$$

By replacing the minus sign with e^{-j180° . Thus,

$$\begin{aligned}
 \Gamma &= 0.76e^{j119.3^\circ}e^{-j180^\circ} \\
 &= 0.76e^{-j60.7^\circ} \\
 &= 0.76\angle -60.7^\circ,
 \end{aligned}$$

or

$$|\Gamma| = 0.76, \quad \theta_r = -60.7^\circ.$$

Example 3:

A $50\ \Omega$ transmission line is terminated in a load with $Z_L = (100 + j50)\ \Omega$. Find the voltage reflection coefficient and the voltage standing-wave ratio.

Solution:

Γ is given by

$$\Gamma = \frac{z_L - 1}{z_L + 1} = \frac{(2 + j1) - 1}{(2 + j1) + 1} = \frac{1 + j1}{3 + j1}.$$

Converting the numerator and denominator to polar form yields

$$\Gamma = \frac{1.414e^{j45^\circ}}{3.162e^{j18.4^\circ}} = 0.45e^{j26.6^\circ}.$$

Using the definition for S , we have

$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.45}{1 - 0.45} = 2.6.$$

Example 4:

A 1.05 GHz generator circuit with series impedance $Z_g = 10\ \Omega$ and voltage source given by

$$v_g(t) = 10 \sin(\omega t + 30^\circ) \quad (\text{V})$$

is connected to a load $Z_L = (100 + j50)\ \Omega$ through a $50\text{-}\Omega$, 67-cm long lossless transmission line. The phase velocity of the line is $0.7c$, where c is the velocity of light in a vacuum. Find $v(d, t)$ and $i(d, t)$ on the line.

Solution:

From the relationship $u_p = \lambda f$, we find the wavelength

$$\lambda = \frac{u_p}{f} = \frac{0.7 \times 3 \times 10^8}{1.05 \times 10^9} = 0.2 \text{ m},$$

and

$$\beta l = \frac{2\pi}{\lambda} l = \frac{2\pi}{0.2} \times 0.67 = 6.7\pi = 0.7\pi = 126^\circ,$$

where we have subtracted multiples of 2π . The voltage reflection coefficient at the load is

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{(100 + j50) - 50}{(100 + j50) + 50} = 0.45e^{j26.6^\circ}.$$

With reference to **Fig 1.25**,

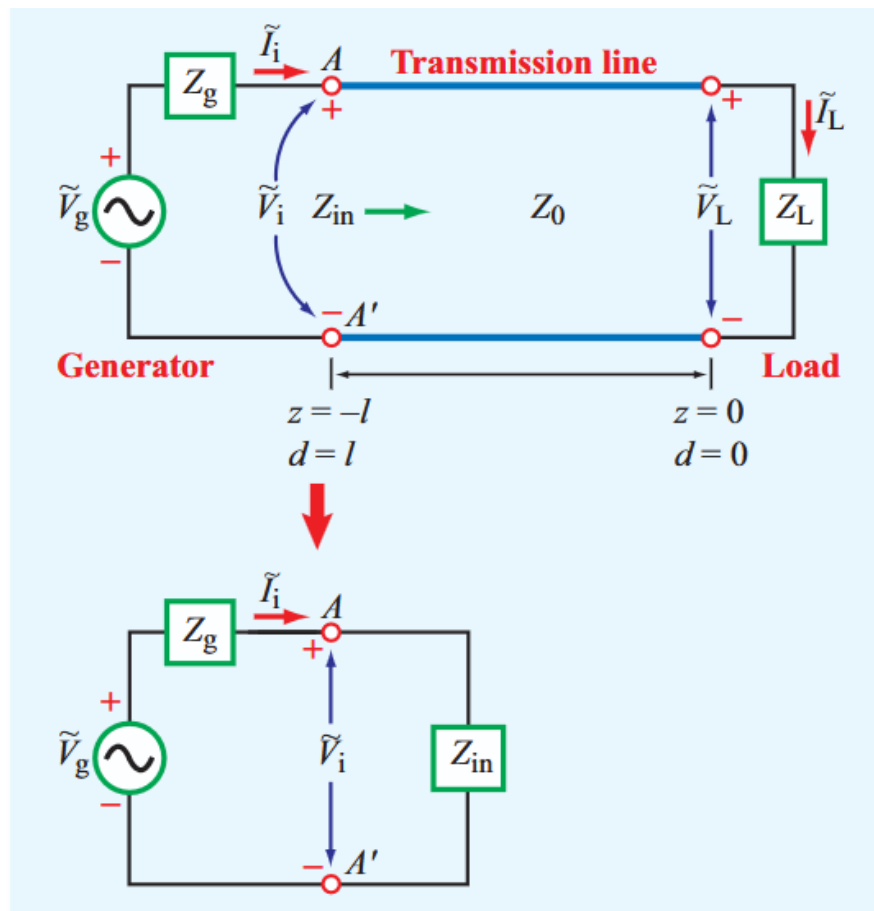


Figure 1-25: At the generator end, the terminated transmission line can be replaced with the input impedance of the line Z_{in}

The input impedance of the line given by

$$\begin{aligned} Z_{in} &= Z_0 \left(\frac{1 + \Gamma_l}{1 - \Gamma_l} \right) \\ &= Z_0 \left(\frac{1 + \Gamma e^{-j2\beta l}}{1 - \Gamma e^{-j2\beta l}} \right) \\ &= 50 \left(\frac{1 + 0.45e^{j26.6^\circ} e^{-j252^\circ}}{1 - 0.45e^{j26.6^\circ} e^{-j252^\circ}} \right) = (21.9 + j17.4) \Omega. \end{aligned}$$

Rewriting the expression for the generator voltage with the cosine reference, we have

$$\begin{aligned}
v_g(t) &= 10 \sin(\omega t + 30^\circ) \\
&= 10 \cos(90^\circ - \omega t - 30^\circ) \\
&= 10 \cos(\omega t - 60^\circ) \\
&= \Re [10e^{-j60^\circ} e^{j\omega t}] = \Re [\tilde{V}_g e^{j\omega t}] \quad (\text{V}).
\end{aligned}$$

Hence, the phasor voltage $\tilde{V}_g$ is given by

$$\tilde{V}_g = 10e^{-j60^\circ} = 10\angle -60^\circ \quad (\text{V}).$$

$$\begin{aligned}
V_0^+ &= \left(\frac{\tilde{V}_g Z_{\text{in}}}{Z_g + Z_{\text{in}}} \right) \left(\frac{1}{e^{j\beta l} + \Gamma e^{-j\beta l}} \right) \\
&= \left[\frac{10e^{-j60^\circ} (21.9 + j17.4)}{10 + 21.9 + j17.4} \right] \\
&\quad \cdot (e^{j126^\circ} + 0.45e^{j26.6^\circ} e^{-j126^\circ})^{-1} \\
&= 10.2e^{j159^\circ} \quad (\text{V}).
\end{aligned}$$

With $z = -d$, the phasor voltage on the line is

$$\begin{aligned}
\tilde{V}(d) &= V_0^+ (e^{j\beta d} + \Gamma e^{-j\beta d}) \\
&= 10.2e^{j159^\circ} (e^{j\beta d} + 0.45e^{j26.6^\circ} e^{-j\beta d}),
\end{aligned}$$

and the corresponding instantaneous voltage $v(d, t)$ is

$$\begin{aligned}
v(d, t) &= \Re [\tilde{V}(d) e^{j\omega t}] \\
&= 10.2 \cos(\omega t + \beta d + 159^\circ) \\
&\quad + 4.55 \cos(\omega t - \beta d + 185.6^\circ) \quad (\text{V}).
\end{aligned}$$

Similarly,

$$\begin{aligned}
\tilde{I}(d) &= 0.20e^{j159^\circ} (e^{j\beta d} - 0.45e^{j26.6^\circ} e^{-j\beta d}), \\
i(d, t) &= 0.20 \cos(\omega t + \beta d + 159^\circ) \\
&\quad + 0.091 \cos(\omega t - \beta d + 185.6^\circ) \quad (\text{A}).
\end{aligned}$$

Example 5:

Choose the length of a shorted $50 \, \Omega$ lossless transmission line (**Fig. 1.26**) such that its input impedance at 2.25 GHz is identical to that of a capacitor with capacitance $C_{\text{eq}} = 4 \, \text{pF}$. The wave velocity on the line is $0.75c$.

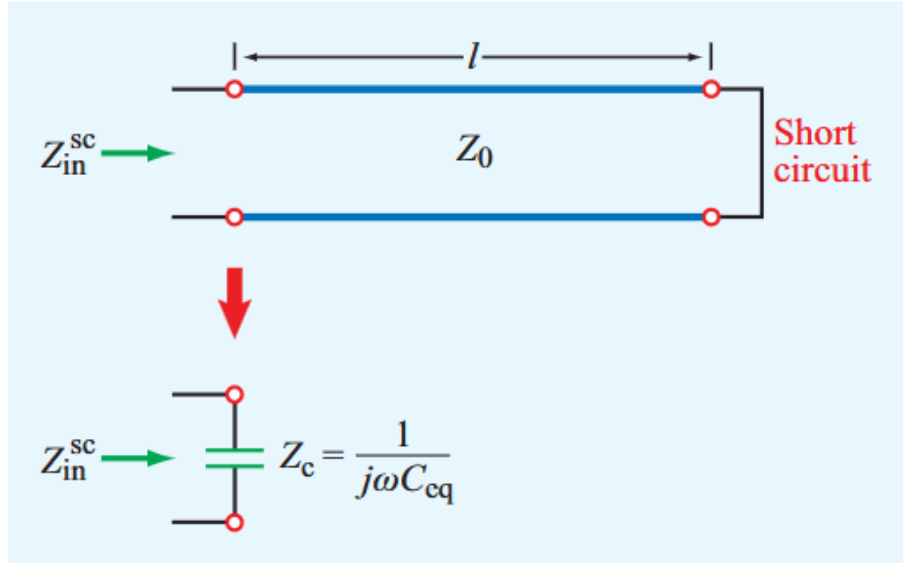


Figure 1-26: Shorted line as an equivalent capacitor

Solution:

We are given

$$\begin{aligned} u_p &= 0.75c = 0.75 \times 3 \times 10^8 = 2.25 \times 10^8 \text{ m/s}, \\ Z_0 &= 50 \, \Omega, \\ f &= 2.25 \text{ GHz} = 2.25 \times 10^9 \text{ Hz}, \\ C_{eq} &= 4 \text{ pF} = 4 \times 10^{-12} \text{ F}. \end{aligned}$$

The phase constant is

$$\beta = \frac{2\pi}{\lambda} = \frac{2\pi f}{u_p} = \frac{2\pi \times 2.25 \times 10^9}{2.25 \times 10^8} = 62.8 \quad (\text{rad/m}).$$

It follows that

$$\begin{aligned} \tan \beta l &= -\frac{1}{Z_0 \omega C_{eq}} \\ &= -\frac{1}{50 \times 2\pi \times 2.25 \times 10^9 \times 4 \times 10^{-12}} = -0.354. \end{aligned}$$

The tangent function is negative when its argument is in the second or fourth quadrants. The solution for the second quadrant is

$$\beta l_1 = 2.8 \text{ rad} \quad \text{or} \quad l_1 = \frac{2.8}{\beta} = \frac{2.8}{62.8} = 4.46 \text{ cm},$$

and the solution for the fourth quadrant is

$$\beta l_2 = 5.94 \text{ rad} \quad \text{or} \quad l_2 = \frac{5.94}{62.8} = 9.46 \text{ cm}.$$

The length l_2 is greater than l_1 by exactly $\lambda/2$. In fact, any length $l = 4.46 \text{ cm} + n\lambda/2$, where n is a positive integer, also is a solution.

Example 6:

An air line has a characteristic impedance of 70Ω and a phase constant of 3 rad/m at 100 MHz . Calculate the inductance per meter and the capacitance per meter of the line.

Solution:

An air line can be regarded as a lossless line because $\sigma \simeq 0$ and $\sigma_c \rightarrow \infty$. Hence

$$R = 0 = G \quad \text{and} \quad \alpha = 0$$

$$Z_o = R_o = \sqrt{\frac{L}{C}} \quad (11.1.1)$$

$$\beta = \omega\sqrt{LC} \quad (11.1.2)$$

$$\frac{R_o}{\beta} = \frac{1}{\omega C}$$

or

$$C = \frac{\beta}{\omega R_o} = \frac{3}{2\pi \times 100 \times 10^6 (70)} = 68.2 \text{ pF/m}$$

$$L = R_o^2 C = (70)^2 (68.2 \times 10^{-12}) = 334.2 \text{ nH/m}$$

Example 7:

A distortionless line has $Z_o = 60 \Omega$, $\alpha = 20 \text{ mNp/m}$, $u = 0.6c$, where c is the speed of light in a vacuum. Find R , L , G , C , and λ at 100 MHz .

Solution:

For a distortionless line,

$$RC = GL \quad \text{or} \quad G = \frac{RC}{L}$$

and hence

$$Z_o = \sqrt{\frac{L}{C}}$$

$$\alpha = \sqrt{RG} = R\sqrt{\frac{C}{L}} = \frac{R}{Z_o}$$

or

$$R = \alpha Z_o$$

But

$$u = \frac{\omega}{\beta} = \frac{1}{\sqrt{LC}}$$

$$R = \alpha Z_o = (20 \times 10^{-3})(60) = 1.2 \Omega/\text{m}$$

$$L = \frac{Z_o}{u} = \frac{60}{0.6 (3 \times 10^8)} = 333 \text{ nH/m}$$

$$G = \frac{\alpha^2}{R} = \frac{400 \times 10^{-6}}{1.2} = 333 \mu\text{S/m}$$

Example 8:

A certain transmission line 2 m long operating at $\omega = 10^6$ rad/s has $\alpha = 8$ dB/m, $\beta = 1$ rad/m, and $Z_o = 60 + j40 \Omega$. If the line is connected to a source of $10\angle 0^\circ$ V, $Z_g = 40 \Omega$ and terminated by a load of $20 + j50 \Omega$, determine

- (a) The input impedance
- (b) The sending-end current
- (c) The current at the middle of the line

Solution:

- (a) Since 1 Np = 8.686 dB,

$$\alpha = \frac{8}{8.686} = 0.921 \text{ Np/m}$$

$$\gamma = \alpha + j\beta = 0.921 + j1 \text{ /m}$$

$$\gamma\ell = 2(0.921 + j1) = 1.84 + j2$$

Using the formula for $\tanh(x + jy)$ in the following,

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}$$

$$\tanh x = \frac{\sinh x}{\cosh x}, \quad \coth x = \frac{1}{\tanh x}$$

$$\tanh(x \pm jy) = \frac{\sinh 2x}{\cosh 2x + \cos 2y} \pm j \frac{\sin 2y}{\cosh 2x + \cos 2y}$$

we obtain

$$\tanh \gamma\ell = 1.033 - j0.03929$$

$$\begin{aligned} Z_{\text{in}} &= Z_o \left(\frac{Z_L + Z_o \tanh \gamma\ell}{Z_o + Z_L \tanh \gamma\ell} \right) \\ &= (60 + j40) \left[\frac{20 + j50 + (60 + j40)(1.033 - j0.03929)}{60 + j40 + (20 + j50)(1.033 - j0.03929)} \right] \\ Z_{\text{in}} &= 60.25 + j38.79 \Omega \end{aligned}$$

- (b) The sending-end current is $I(z = 0) = I_o$.

$$\begin{aligned} I(z = 0) &= \frac{V_g}{Z_{\text{in}} + Z_g} = \frac{10}{60.25 + j38.79 + 40} \\ &= 93.03\angle -21.15^\circ \text{ mA} \end{aligned}$$

- (c) To find the current at any point, we need V_o^+ and V_o^- . But

$$I_o = I(z = 0) = 93.03\angle -21.15^\circ \text{ mA}$$

$$V_o = Z_{\text{in}} I_o = (71.66\angle 32.77^\circ)(0.09303\angle -21.15^\circ) = 6.667\angle 11.62^\circ \text{ V}$$

$$\begin{aligned}
V_o^+ &= \frac{1}{2}(V_o + Z_o I_o) \\
&= \frac{1}{2} [6.667 \angle 11.62^\circ + (60 + j40)(0.09303 \angle -21.15^\circ)] = 6.687 \angle 12.08^\circ
\end{aligned}$$

$$V_o^- = \frac{1}{2}(V_o - Z_o I_o) = 0.0518 \angle 260^\circ$$

At the middle of the line, $z = \ell/2$, $\gamma z = 0.921 + j1$. Hence, the current at this point is

$$\begin{aligned}
I_s(z = \ell/2) &= \frac{V_o^+}{Z_o} e^{-\gamma z} - \frac{V_o^-}{Z_o} e^{\gamma z} \\
&= \frac{(6.687 e^{j12.08^\circ}) e^{-0.921 - j1}}{60 + j40} - \frac{(0.0518 e^{j260^\circ}) e^{0.921 + j1}}{60 + j40}
\end{aligned}$$

Note that $j1$ is in radians and is equivalent to $j57.3^\circ$. Thus,

$$\begin{aligned}
I_s(z = \ell/2) &= \frac{6.687 e^{j12.08^\circ} e^{-0.921} e^{-j57.3^\circ}}{72.1 e^{j33.69^\circ}} - \frac{0.0518 e^{j260^\circ} e^{0.921} e^{j57.3^\circ}}{72.1 e^{j33.69^\circ}} \\
&= 0.0369 e^{-j78.91^\circ} - 0.001805 e^{j283.61^\circ} \\
&= 6.673 - j34.456 \text{ mA} \\
&= 35.10 \angle 281^\circ \text{ mA}
\end{aligned}$$

Example 9:

A lossless transmission line with $Z_o = 50 \Omega$ is 30 m long and operates at 2 MHz. The line is terminated with a load $Z_L = 60 + j40 \Omega$. If $u = 0.6c$ on the line, find

- (a) The reflection coefficient Γ
- (b) The standing wave ratio s
- (c) The input impedance

Solution:

(a)

$$\begin{aligned}
\Gamma &= \frac{Z_L - Z_o}{Z_L + Z_o} = \frac{60 + j40 - 50}{60 + j40 + 50} = \frac{10 + j40}{110 + j40} \\
&= 0.3523 \angle 56^\circ
\end{aligned}$$

(b)

$$s = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.3523}{1 - 0.3523} = 2.088$$

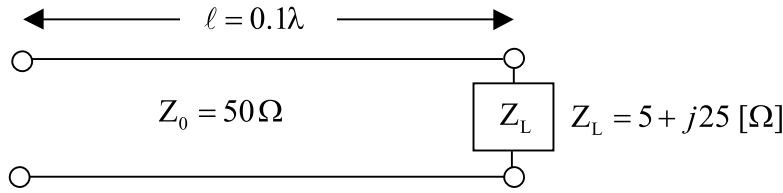
(c) Since $u = \omega/\beta$, or $\beta = \omega/u$,

$$\beta \ell = \frac{\omega \ell}{u} = \frac{2\pi(2 \times 10^6)(30)}{0.6(3 \times 10^8)} = \frac{2\pi}{3} = 120^\circ$$

Note that $\beta\ell$ is the electrical length of the line.

$$\begin{aligned}
 Z_{\text{in}} &= Z_o \left[\frac{Z_L + jZ_o \tan \beta\ell}{Z_o + jZ_L \tan \beta\ell} \right] \\
 &= \frac{50(60 + j40 + j50 \tan 120^\circ)}{[50 + j(60 + j40) \tan 120^\circ]} \\
 &= \frac{50(6 + j4 - j5\sqrt{3})}{(5 + 4\sqrt{3} - j6\sqrt{3})} = 24.01 \angle 3.22^\circ \\
 &= 23.97 + j1.35 \, \Omega
 \end{aligned}$$

Smith Chart Problems



1. The 0.1λ length line shown has a characteristic impedance of 50Ω and is terminated with a load impedance of $Z_L = 5 + j25\Omega$.

- (a) Locate $z_L = \frac{Z_L}{Z_0} = 0.1 + j0.5$ on the Smith chart.

See the point plotted on the Smith chart.

- (b) What is the impedance at $\ell = 0.1\lambda$?

Since we want to move away from the load (i.e., toward the generator), read 0.074λ on the WAVELENGTHS TOWARD GENERATOR scale and add $\ell = 0.1\lambda$ to obtain 0.174λ on the WAVELENGTHS TOWARD GENERATOR scale. A radial line from the center of the chart intersects the constant reflection coefficient magnitude circle at $z = 0.38 + j1.88$. Hence $Z = zZ_0 = 50(0.38 + j1.88) = 19 + j94\Omega$.

- (c) What is the VSWR on the line?

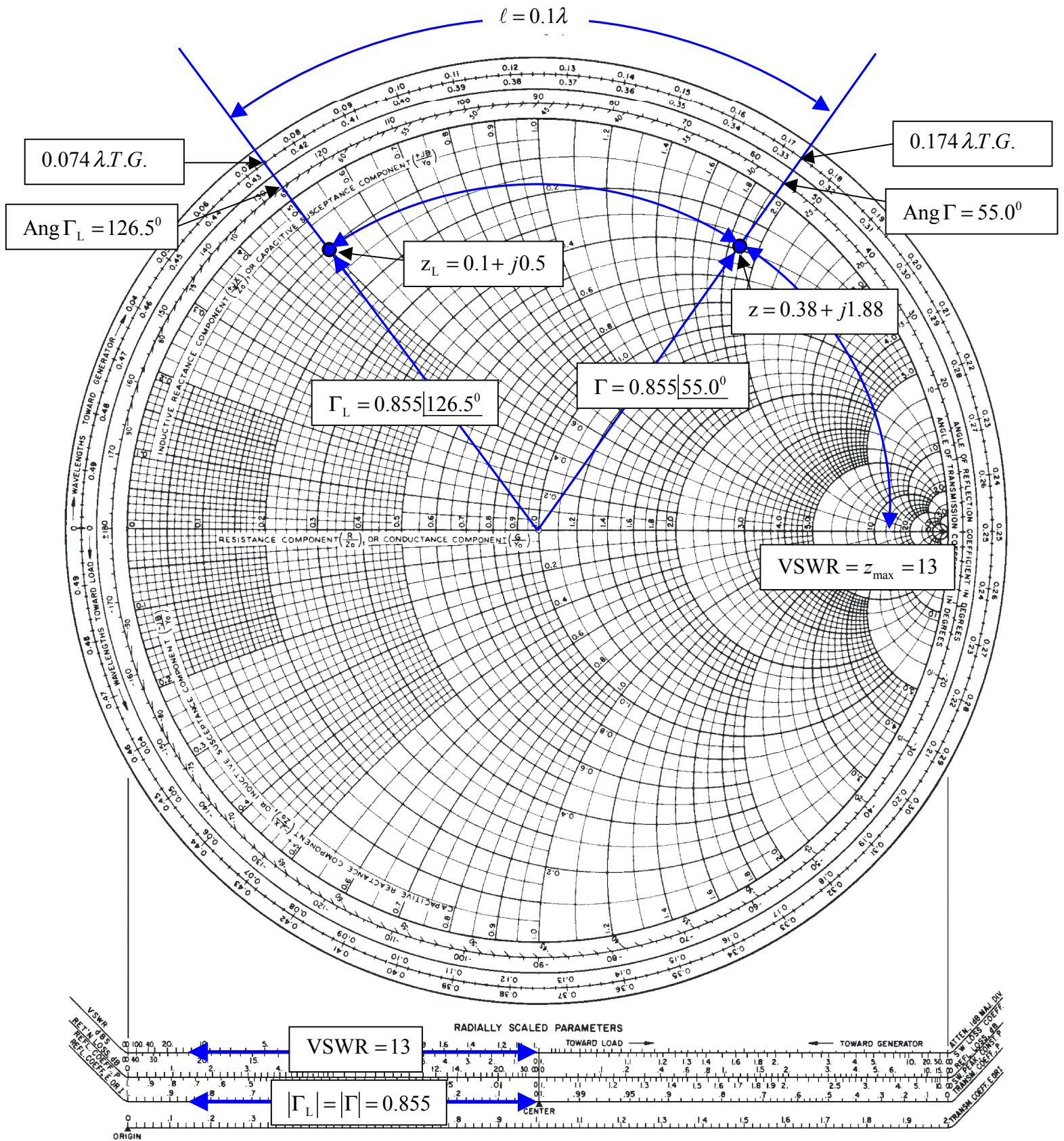
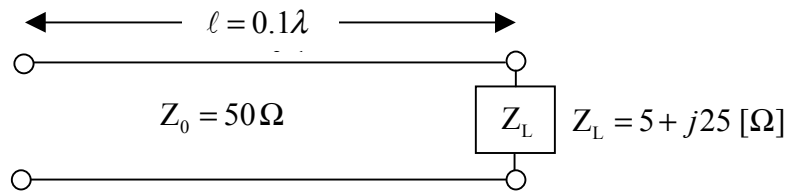
Find $VSWR = z_{\max} = 13$ on the horizontal line to the right of the chart's center. Or use the SWR scale on the chart.

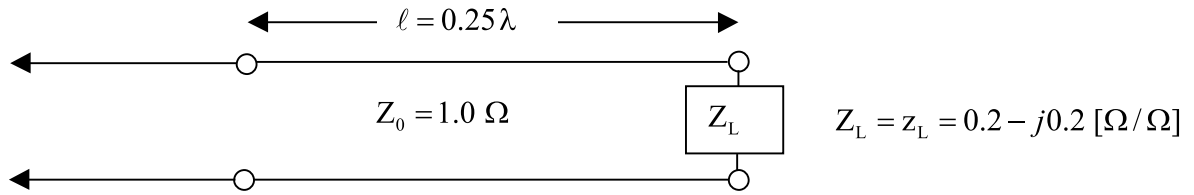
- (d) What is Γ_L ?

From the REFLECTION COEFFICIENT scale below the chart, find $|\Gamma_L| = 0.855$. From the ANGLE OF REFLECTION COEFFICIENT scale on the perimeter of the chart, find the angle of $\Gamma_L = 126.5^\circ$. Hence $\Gamma_L = 0.855e^{j126.5^\circ}$.

- (e) What is Γ at $\ell = 0.1\lambda$ from the load?

Note that $|\Gamma| = |\Gamma_L| = 0.855$. Read the angle of the reflection coefficient from the ANGLE OF REFLECTION COEFFICIENT scale as 55.0° . Hence $\Gamma = 0.855e^{j55.0^\circ}$.





2. A transmission line has $Z_0 = 1.0$, $Z_L = z_L = 0.2 - j0.2\Omega$.

- (a) What is z at $\ell = \frac{\lambda}{4} = 0.25\lambda$?

From the chart, read 0.467λ from the WAVELENGTHS TOWARD GENERATOR SCALE. Add 0.25λ to obtain 0.717λ on the WAVELENGTHS TOWARD GENERATOR SCALE. This is not on the chart, but since it repeats every half wavelength, it is the same as $0.717\lambda - 0.500\lambda = 0.217\lambda$. Drawing a radial line from the center of the chart, we find an intersection with the constant reflection coefficient magnitude circle at $z = Z = 2.5 + j2.5$.

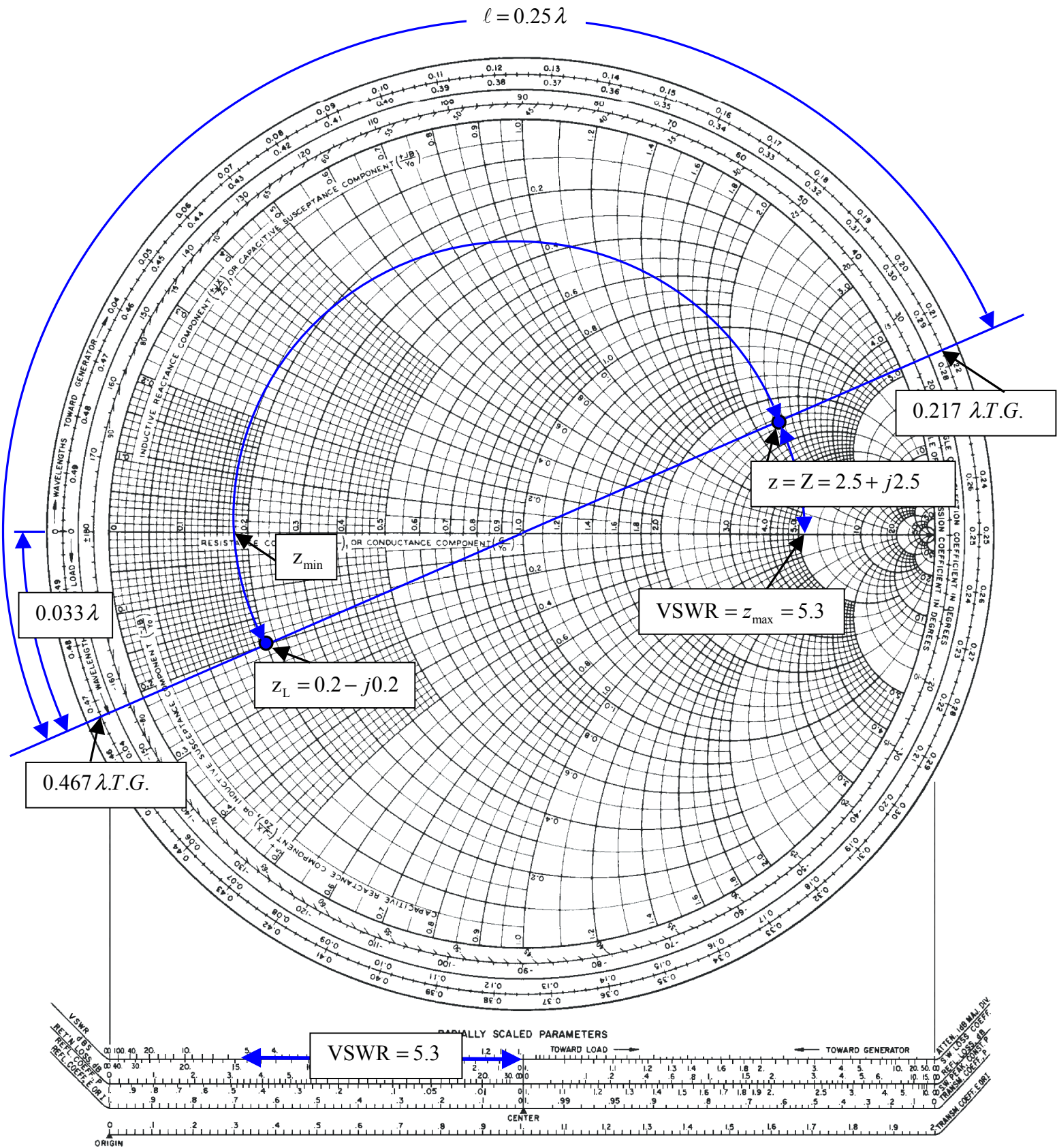
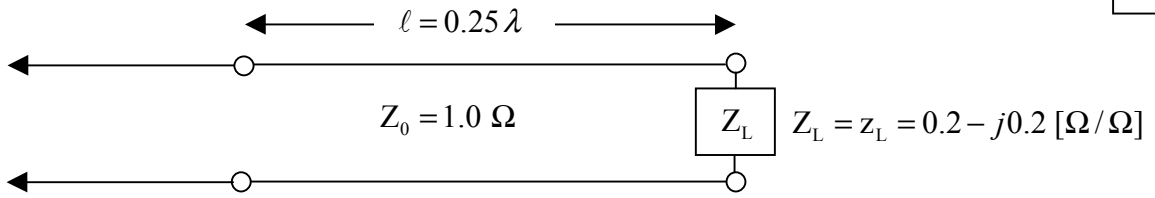
- (b) What is the VSWR on the line?

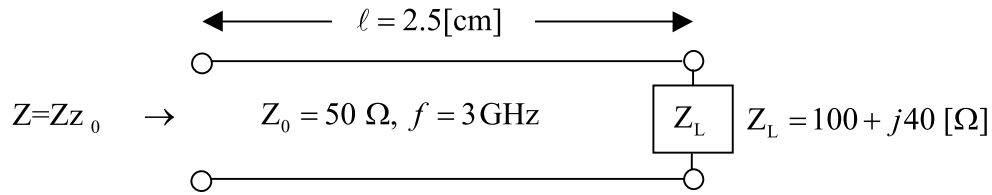
From the intersection of the constant reflection coefficient circle with the right hand side of the horizontal axis, read $\text{VSWR} = z_{\max} = 5.3$.

- (c) How far from the load is the first voltage minimum? The first current minimum?

The voltage minimum occurs at $z_{\min}$ which is at a distance of $0.500\lambda - 0.467\lambda = 0.033\lambda$ from the load. Or read this distance directly on the WAVELENGTHS TOWARD LOAD scale.

The current minimum occurs at $z_{\max}$ which is a quarter of a wavelength farther down the line or at $0.033\lambda + 0.25\lambda = 0.283\lambda$ from the load.





3. The air-filled two-wire line has a characteristic impedance of 50Ω and is operated at $f = 3 \text{ GHz}$. The load is $Z_L = 100 + j40\Omega$.

(a) For the line above, find z_L on the chart.

$$\text{The normalized load is } z_L = \frac{Z_L}{Z_0} = \frac{100 + j40}{50} = 2.0 + j0.8.$$

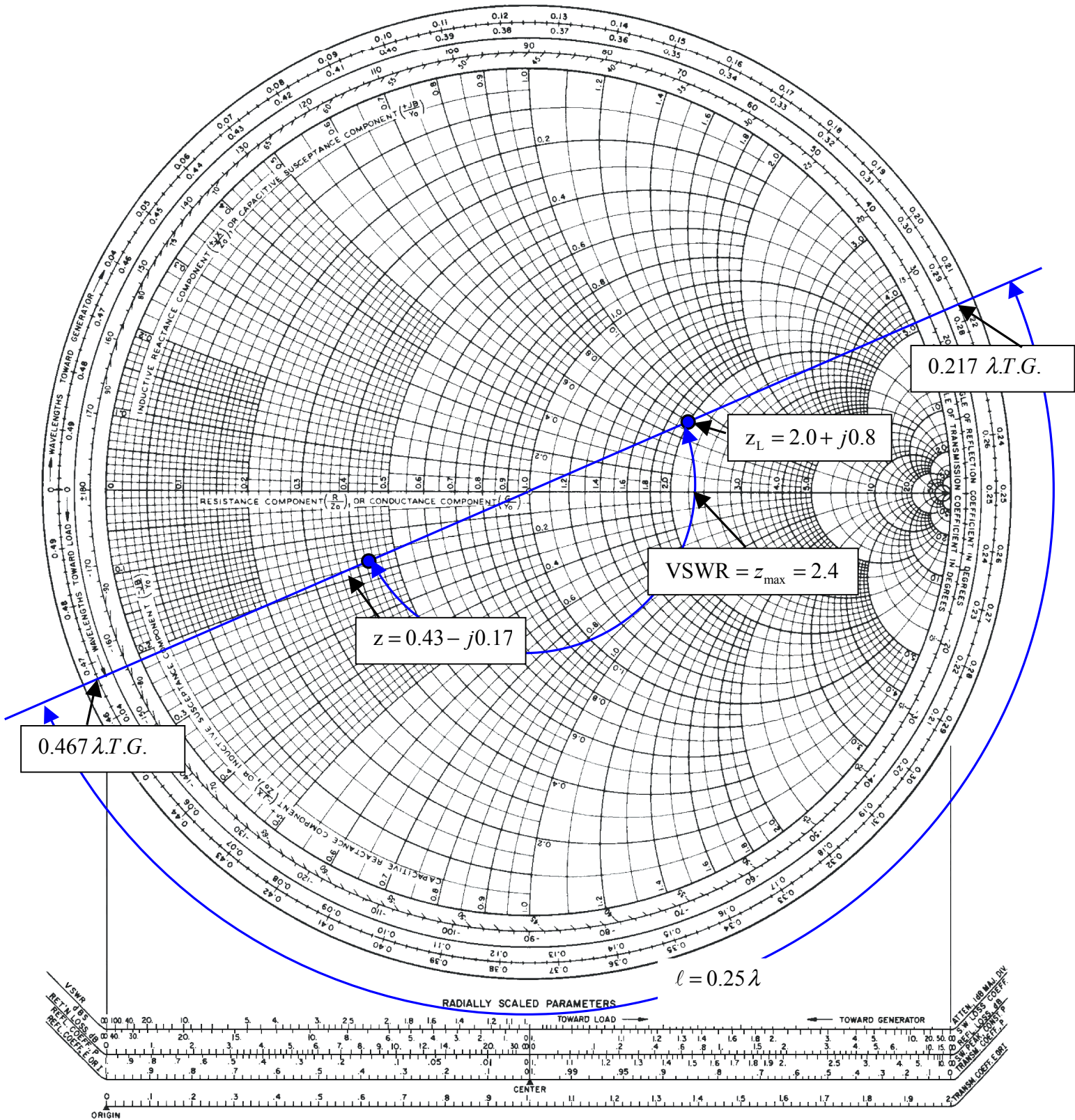
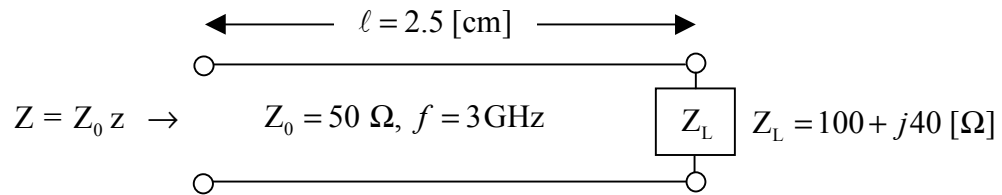
See the Smith chart for location of point.

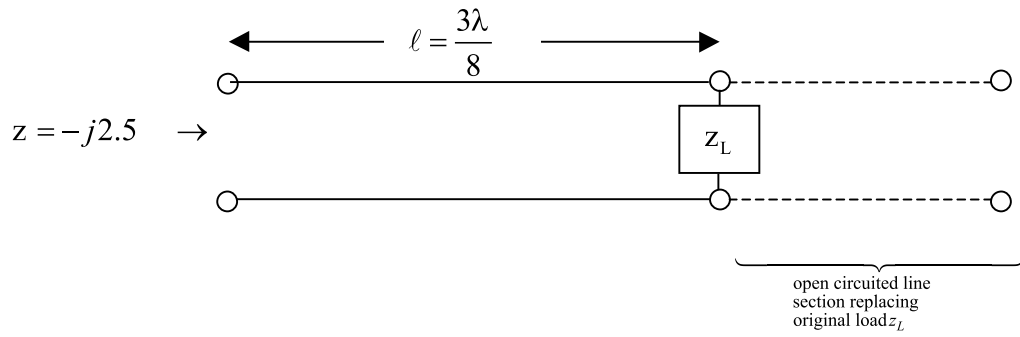
(b) What is the line impedance 2.5 cm from the load?

Note that $\lambda = \frac{c}{f} = \frac{3 \times 10^{10} \text{ cm/sec}}{3 \times 10^9 \text{ Hz}} = 10 \text{ cm}$. Since we are going to move toward the generator (away from the load), at the normalized load position, first read 0.217λ on the WAVELENGTHS TOWARD GENERATOR scale. Then add $2.5 \text{ cm}/10 \text{ cm} = 0.25\lambda$ to this value to obtain 0.467λ on the WAVELENGTHS TOWARD GENERATOR scale. A radial line from the center at this point intersects the constant reflection coefficient magnitude circle at $z = 0.43 - j0.17$, so $Z = zZ_0 = 50(0.43 + j0.17) = 21.5 - j8.5\Omega$.

(c) What is the VSWR on the line?

From the intersection of the reflection coefficient circle and the horizontal axis on the right hand side of the chart, read $\text{VSWR} = 2.4$. Or use the SWR scale below the chart.





4. The line shown is $3/8\lambda$ long and its normalized input impedance is $z = -j2.5$.

(a) What is the normalized receiving or load end impedance, z_L ?

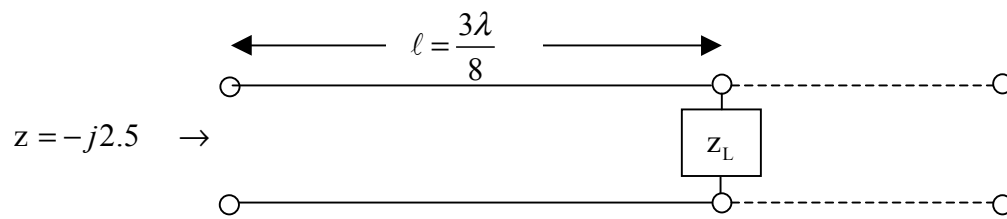
Since we must move toward the load to find the load impedance, locate $z = -j2.5$ on the chart and find the intersection on the WAVELENGTHS TOWARD LOAD scale at 0.189λ . Add the distance $3/8\lambda = 0.375\lambda$ to get 0.064λ on the WAVELENGTHS TOWARD LOAD scale. At the intersection of the line from the center of the chart and the constant reflection coefficient circle, read $z_L = -j0.425$.

(b) What is the distance from the load to the first voltage minimum?

The voltage minimum occurs at the impedance minimum on the horizontal line to the left of the chart's center. This distance can be read directly off the WAVELENGTHS TOWARD LOAD scale (why?) as 0.064λ . Note that the line impedance at the voltage minimum would be that of a short circuit.

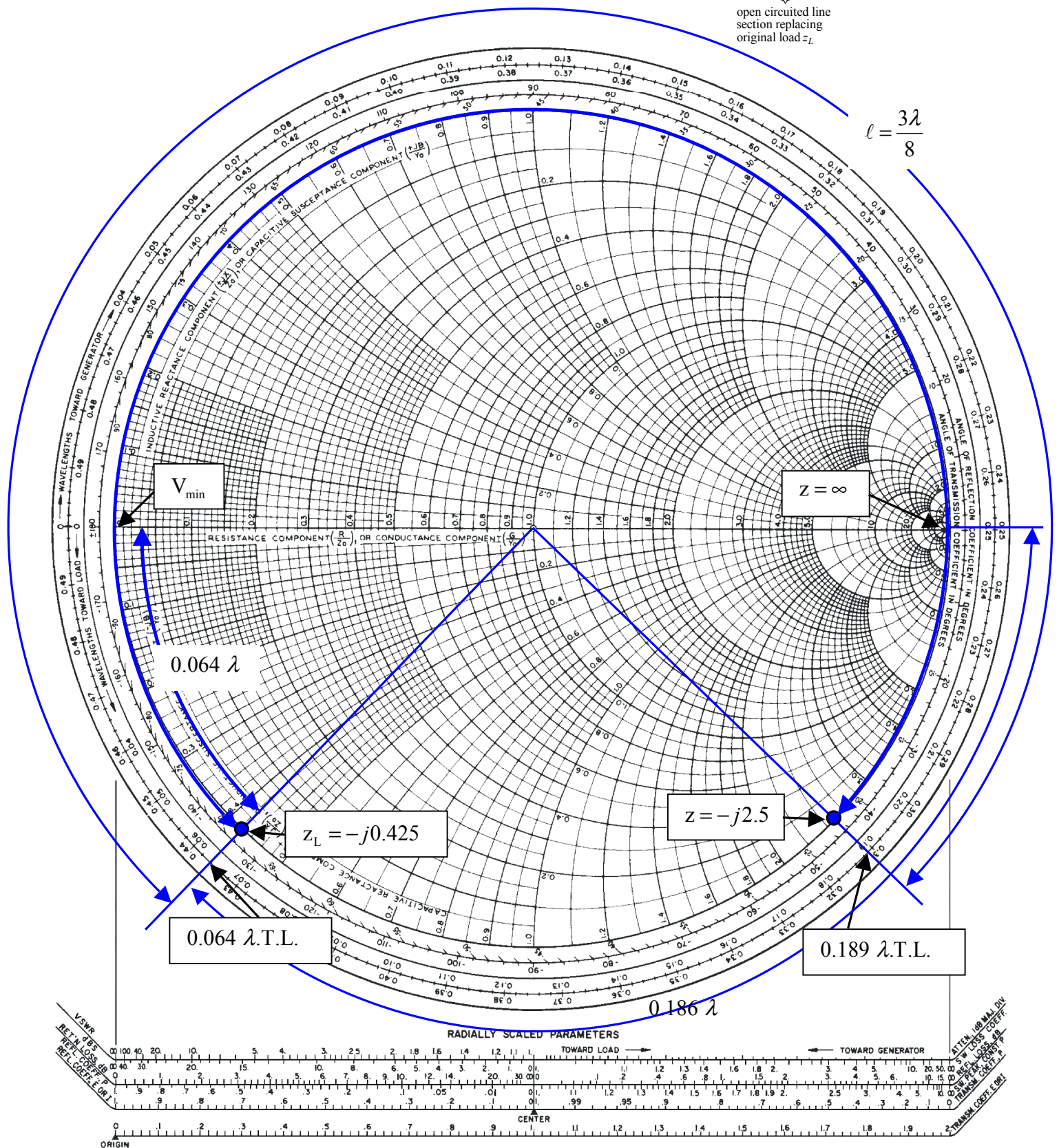
(c) What length of open-circuited line could be used to replace z_L ?

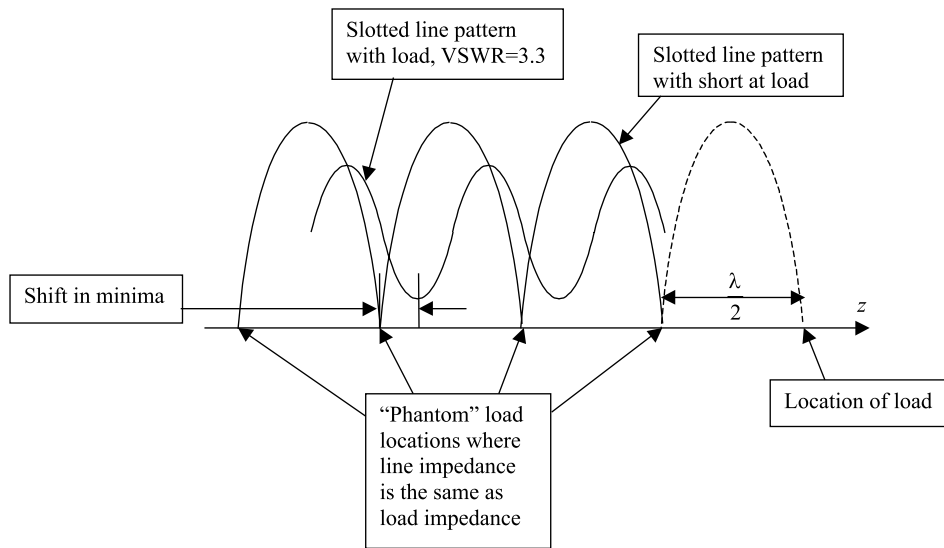
From the position of z_L , we want to move toward the new load (i.e., toward the end of the open-circuited line) until we reach the open circuit condition $z = \infty$. This occurs at 0.25λ on the WAVELENGTHS TOWARD LOAD scale and the total distance moved is hence $0.25\lambda - 0.064\lambda = 0.186\lambda$, which is the length of open-circuited line needed to replace z_L .



open circuited line
section replacing
original load z_L

$$\ell = \frac{3\lambda}{8}$$



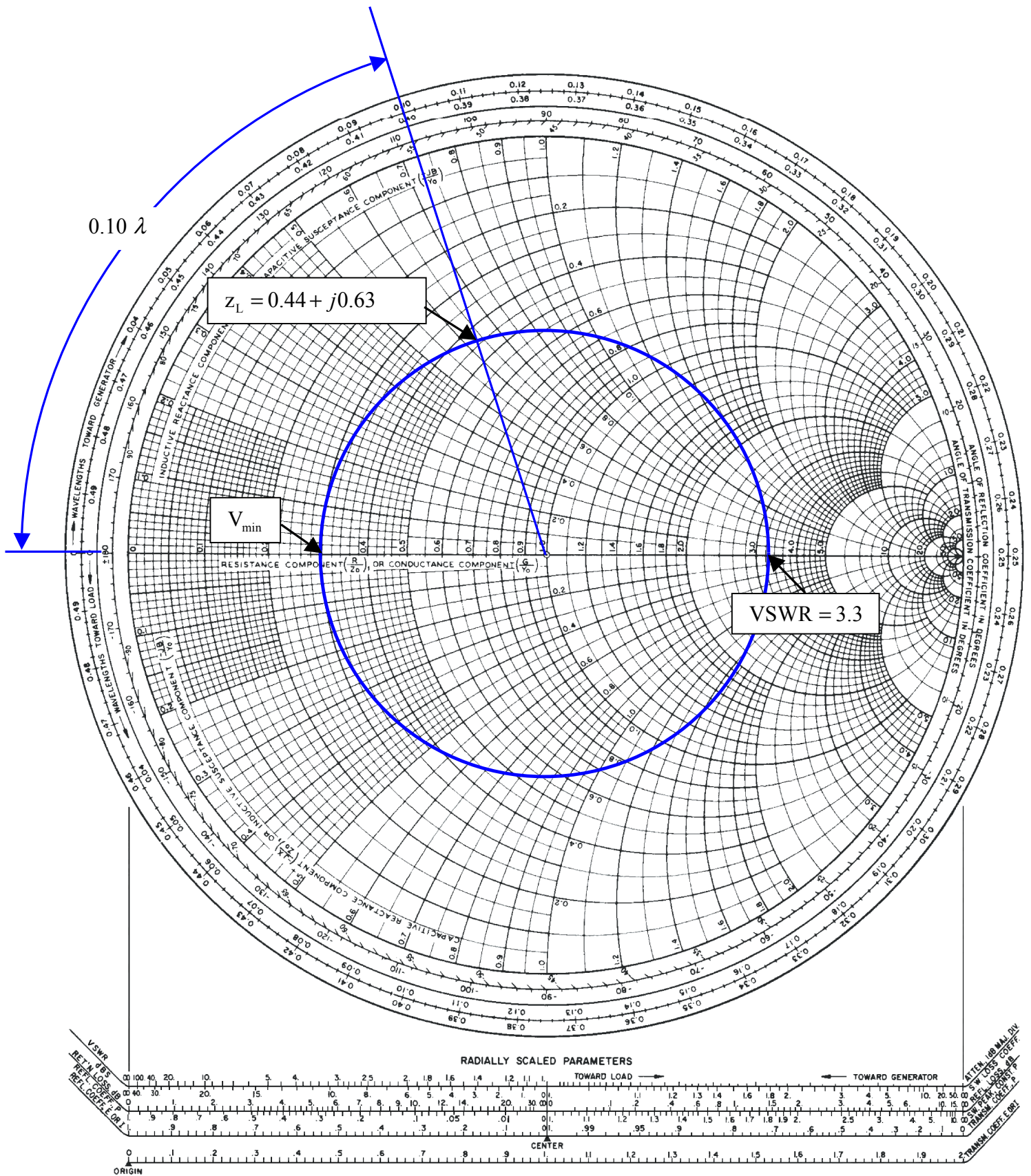


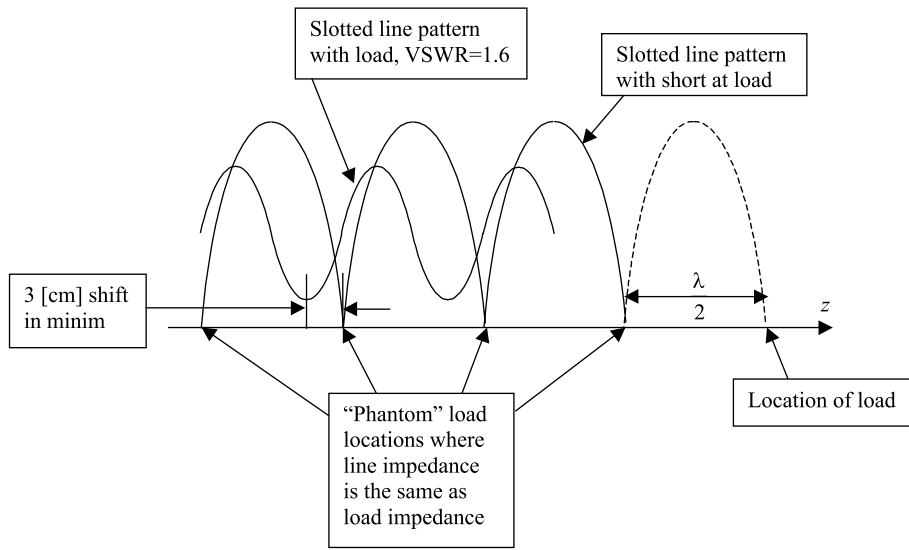
5. An air-filled 50Ω coaxial line has a loaded VSWR of 3.3 at a frequency of 3GHz. Replacing the load with a short causes the voltage minimum to move 1.0 cm towards the generator. What is the normalized load impedance?

For an air-filled line, the wavelength is

$$\lambda = \frac{c}{f} = \frac{3 \times 10^{10} \text{ cm/sec}}{3 \times 10^9 \text{ Hz}} = 10 \text{ cm.}$$

With the VSWR located at $z_{\max} = 3.3$, draw the circle of constant reflection coefficient magnitude. With the load in place, our only information is the location of a voltage minimum (equivalent to an impedance minimum). But thinking of the Smith chart as a crank diagram, we know that the voltage minimum must be at the intersection of the constant reflection coefficient circle and the horizontal axis on the left hand side of the chart between $z = 0$ and $z = 1.0$, (i.e., at $z_{\min} = 0.3$). With this information, we have thus established a correspondence between a location on the line and a point on the chart. Now, move from this minimum voltage point on the chart $\frac{1.0 \text{ cm}}{10.0 \text{ cm}} = 0.1\lambda$ toward the generator to the location where the voltage minimum was measured in the shorted line measurement. Since it was obtained from a shorted line measurement, the location of this point must be an integral number of half-wavelengths from the load (and hence must have the same line impedance as at the load). The normalized impedance here is therefore $z_L = 0.44 + j0.63$. For a line with a 50Ω characteristic impedance, the load impedance is thus $Z_L = 50(0.44 + j0.63) = 22 + j31.5$.





6. A slotted line measurement on an air-filled TEM line yields a VSWR of 1.6 at a frequency of 1 GHz. When the load is replaced by a short, the voltage minimum moves 3 cm towards the load. Find the normalized load impedance.

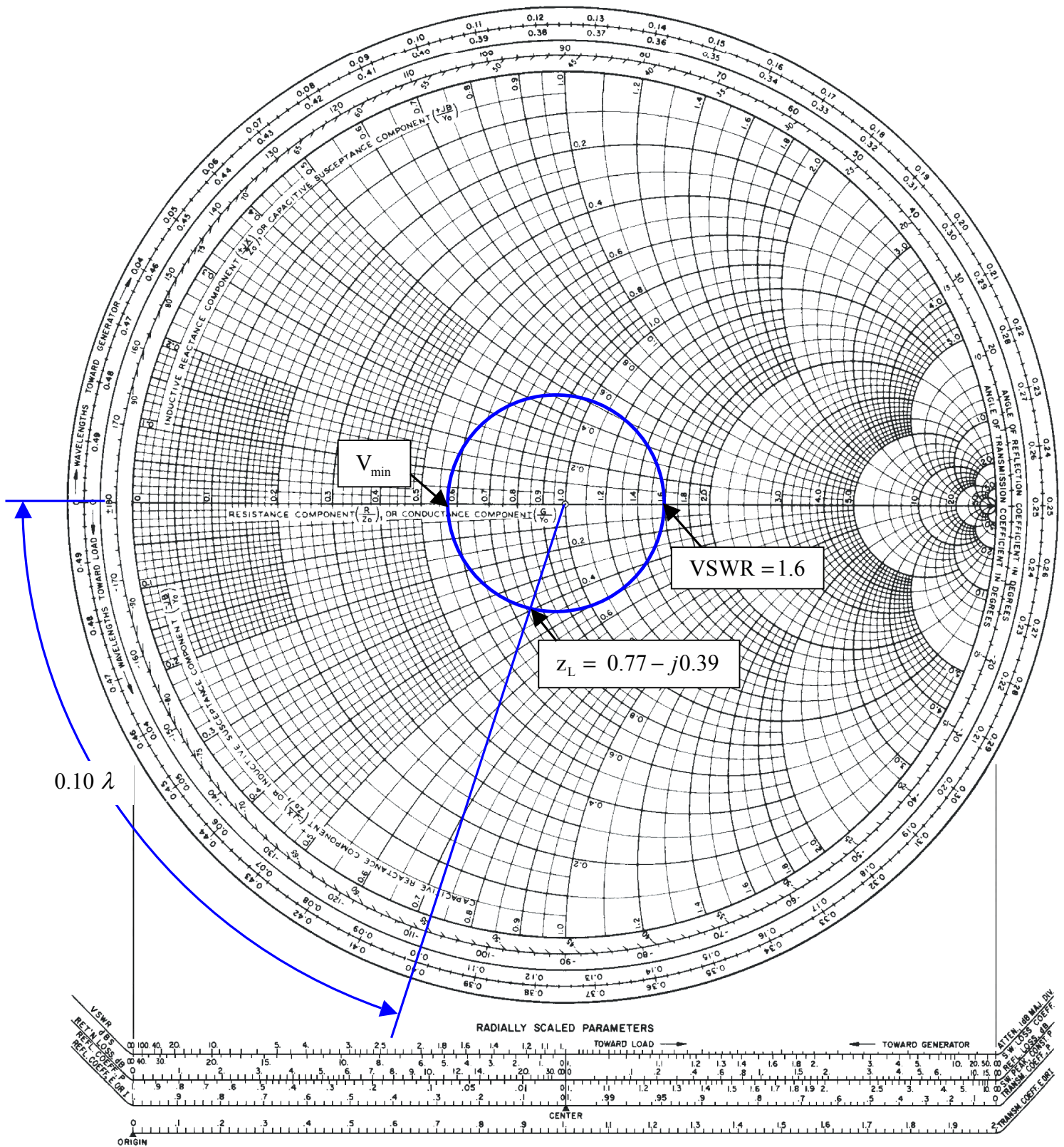
Locate the reflection coefficient circle by its intersection with $z_{\max} = \text{VSWR} = 1.6$. The wavelength for an air-filled TEM line is

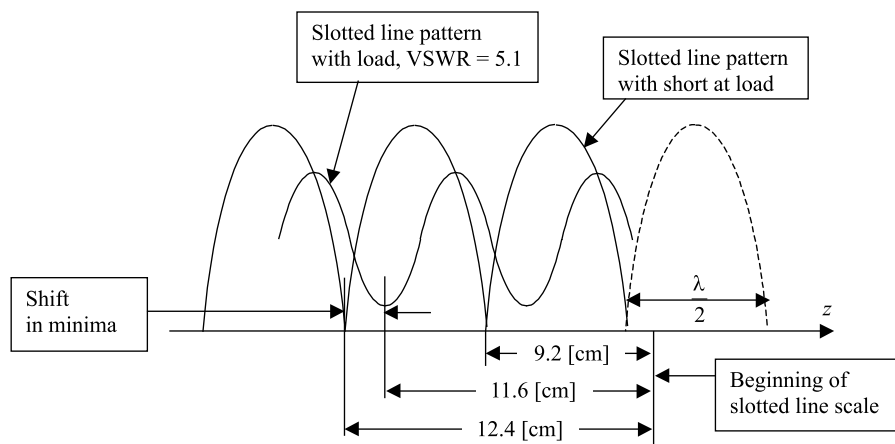
$$\lambda = c/f = \frac{3 \times 10^{10} \text{ cm/s}}{10^9 \text{ Hz}} = 30 \text{ cm}.$$

Hence the shift in the voltage minimum is

$$\frac{3 \text{ cm}}{30 \text{ cm}/\lambda} = 0.1\lambda.$$

Locate the voltage minimum at the impedance minimum, $z_{\min} = 0.625$ and move 0.1λ toward the load to a position an integral number of half-wavelengths from the load. The impedance there, $z_L = 0.77 - j0.39$, is the same as that of the load.





7. A slotted line measurement yields the following parameter values:
 - (a) Voltage minima at 9.2 cm and 12.4 cm measured away from the load with the line terminated in a short.
 - (b) VSWR = 5.1 with the line terminated in the unknown load; a voltage minimum is located 11.6 cm measured away from the load.

What is the normalized line impedance?

Note that this data could have come from either a waveguide or a TEM line measurement. If the transmission system is a waveguide, then the wavelength used is actually the *guide* wavelength. From the voltage minima on the shorted line, the (guide) wavelength may be determined:

$$\frac{\lambda_g}{2} = 12.4 \text{ cm} - 9.2 \text{ cm} = 3.2 \text{ cm}$$

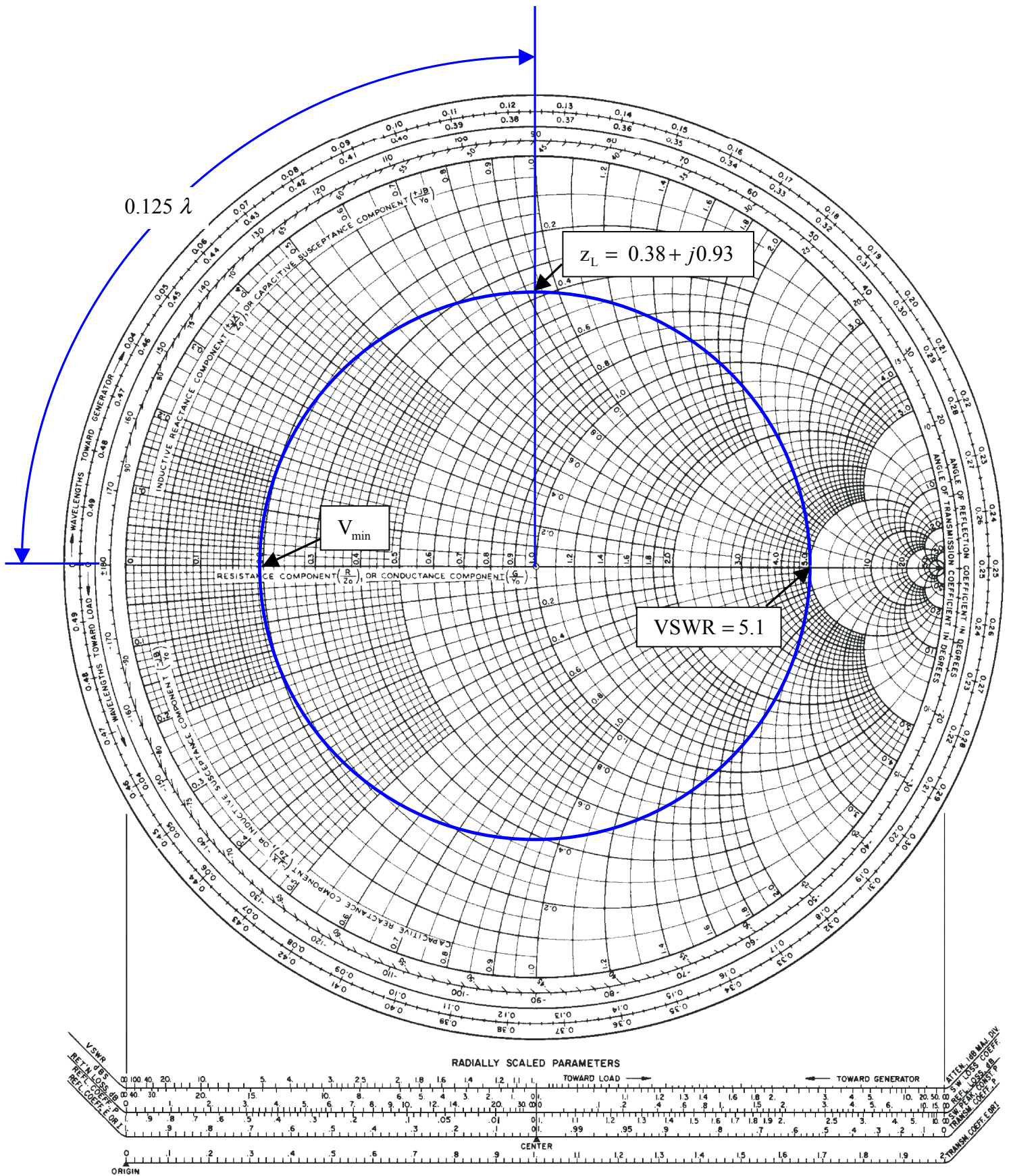
or

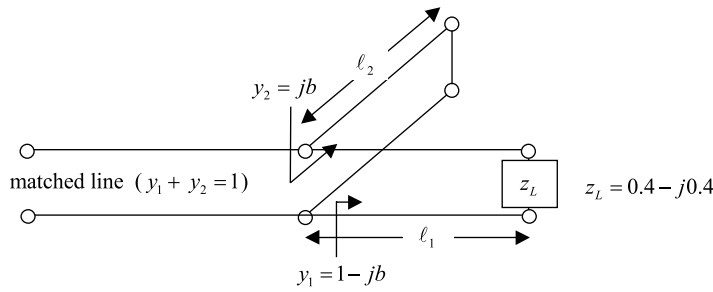
$$\lambda_q = 6.4 \text{ cm.}$$

Hence the shift in the voltage minimum when the load is replaced by a short is

$$\frac{12.4 \text{ cm} - 11.6 \text{ cm}}{6.4 \text{ cm}/\lambda_q} = 0.125\lambda_g$$

toward the generator. Locate the reflection coefficient magnitude circle by its intersection with $z_{\max} = \text{VSWR} = 5.1$ on the horizontal axis. Then from the voltage minimum opposite $z_{\max}$, move $0.125\lambda_g$ toward the generator to find a position an integral number of half-wavelengths from the load. The impedance there is the same as that of the load, $z_L = 0.38 + j0.93$. Alternatively, move $0.5\lambda_g - 0.125\lambda_g = 0.375\lambda_g$ toward the load to locate the same value.





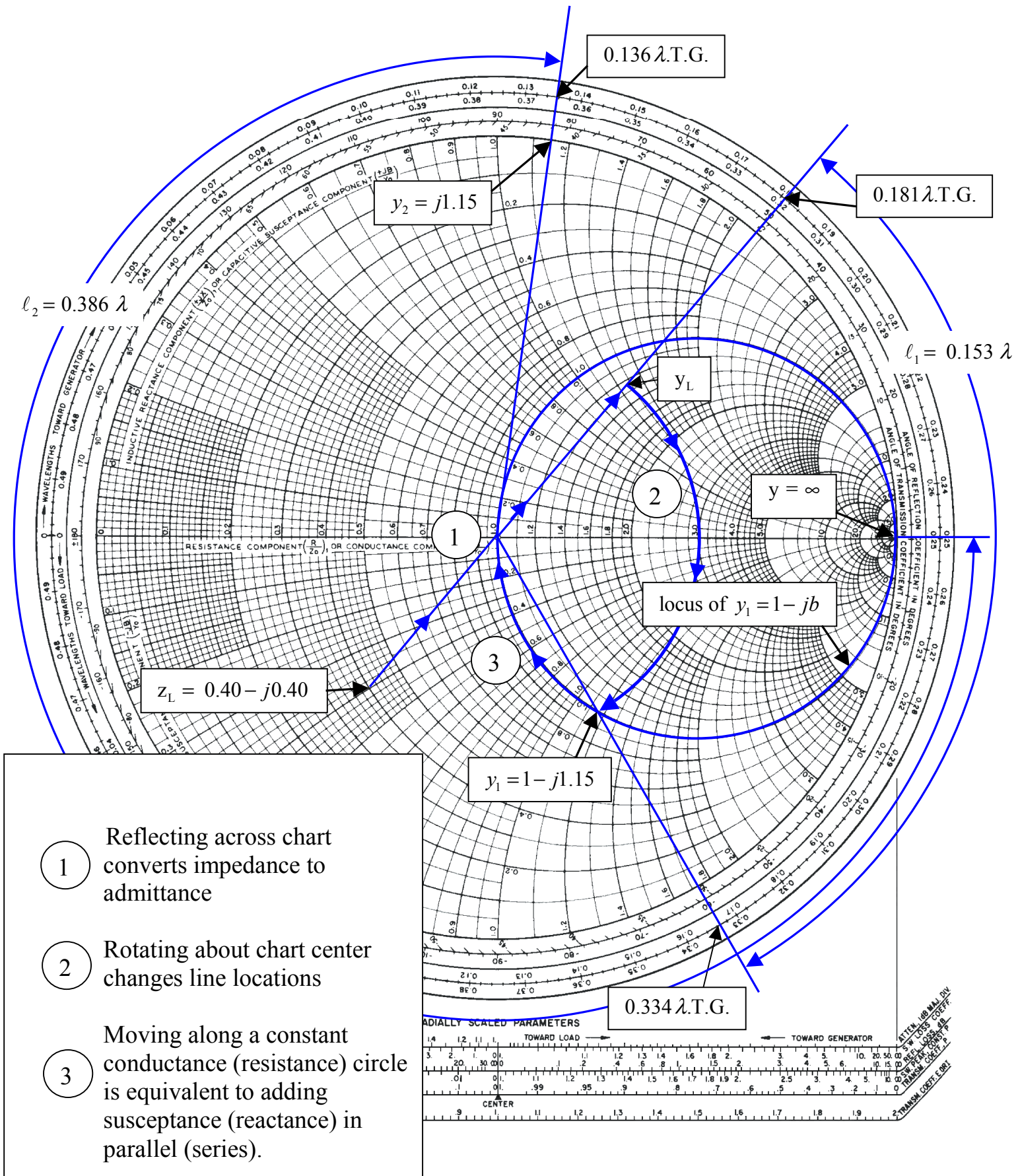
8. Use a single parallel stub tuner to match the above line to its load. Use a shorted stub and find its distance from the load, ℓ_1 , and its length ℓ_2 .

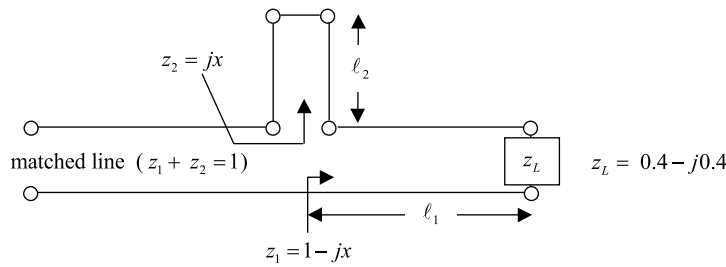
First, locate the load impedance $z_L = 0.4 - j0.4$ on the chart and reflect it to obtain the load admittance, $y_L = 1.24 + j1.24$. To match the line, the parallel combination of the load and stub line sections must present a unit (normalized) admittance at the junction point, i.e. $y_1 + y_2 = 1.0$. Since the stub section is lossless, its input admittance must be purely susceptive, i.e. of the form $y_2 = jb$, and hence the input admittance of the loaded section must have the form $y_1 = 1.0 - jb$. The locus of y_1 is thus the unit conductance circle.

To determine the unknown susceptive part of y_1 , we move from the point on the chart corresponding to y_L toward the generator until we reach the intersection with the unit conductance circle, yielding $y_1 = 1.0 - j1.15$ and thus $y_2 = j1.15$. The electrical distance moved between the points y_L and y_1 is $\ell_1 = 0.334\lambda - 0.181\lambda = 0.153\lambda$ measured on the WAVELENGTHS TOWARD GENERATOR scale.

To find ℓ_2 , locate $y_2 = j1.15$ and move toward the load to the infinite admittance point on the chart, corresponding to a shorted stub. The shorted stub length is therefore $\ell_2 = 0.386\lambda$, the sum of 0.25λ measured on the WAVELENGTHS TOWARD LOAD scale and 0.136λ measured on the WAVELENGTHS TOWARD GENERATOR scale (Why two different scales?).

Note that a second solution may be obtained by rotating past the first intersection to the second intersection with the unit conductance circle at $y_1 = 1.0 + j1.15$. This will yield a longer section of line ℓ_1 . Also, we will have $y_2 = -j1.15$ and the corresponding length of the shorted stub section ℓ_2 will be shorter than in the previous solution. In this case it is not clear which solution is preferable. In general, however, if one solution results in *both* ℓ_1 and ℓ_2 being shorter, then that configuration normally has the greater bandwidth.





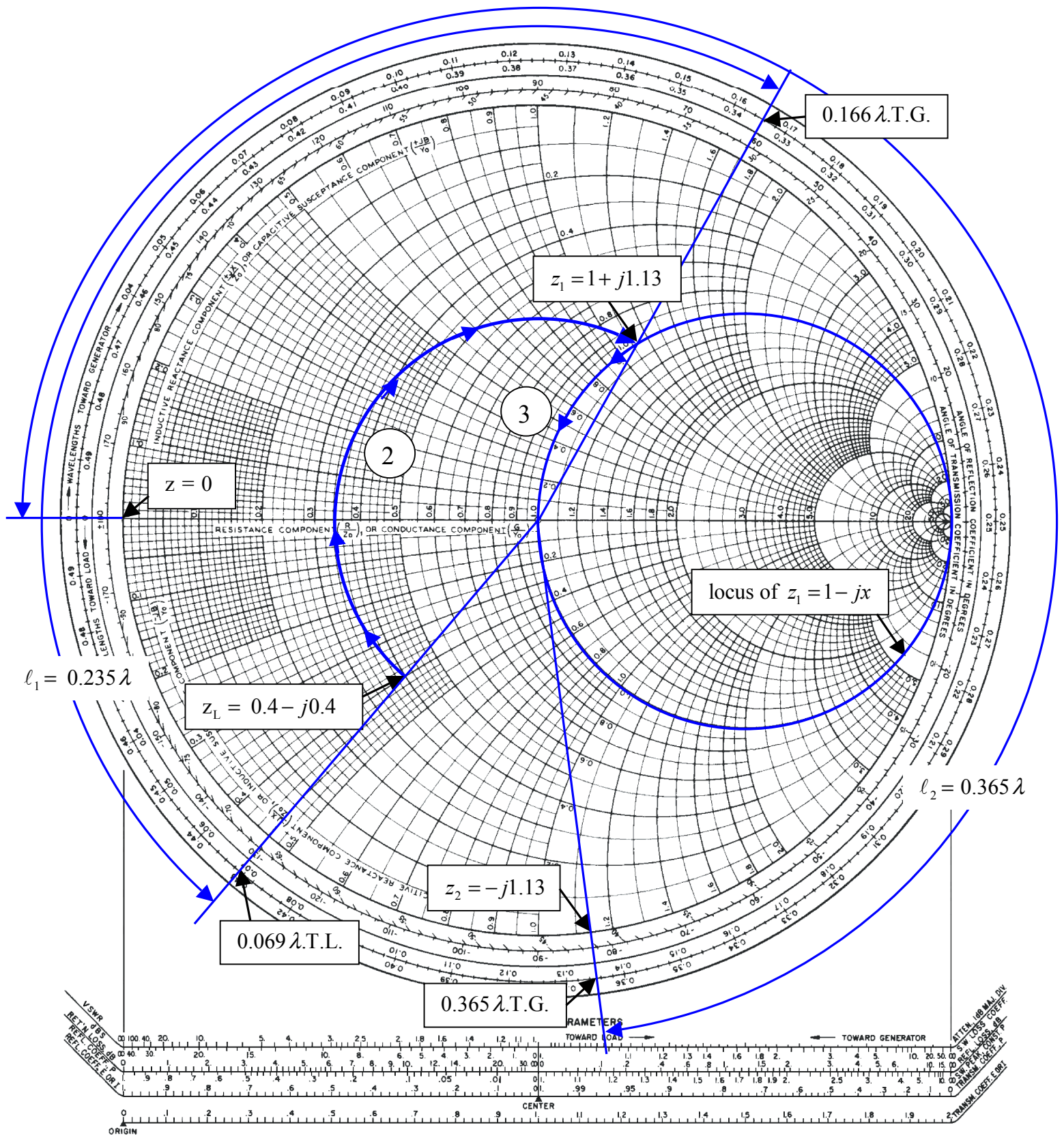
9. Use a single series stub tuner to match the above line to its normalized load. Use a shorted stub and find its distance from the load, ℓ_1 , and its length ℓ_2 .

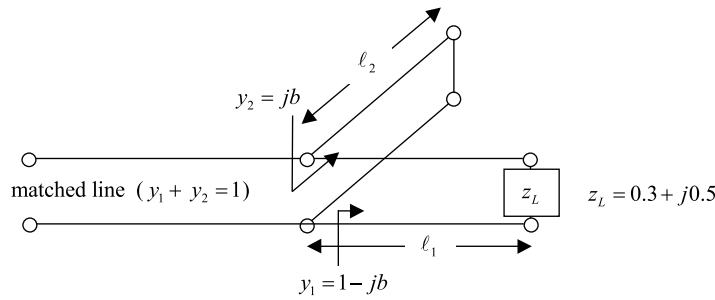
This problem is similar to the parallel stub tuner except that, since the stub is in series with the line, constructions are done using impedances rather than admittances. First, locate the load impedance $z_L = 0.4 - j0.4$ on the chart. To match the line, the series combination of the load and stub line sections must present a unit (normalized) impedance at the junction point, i.e. $z_1 + z_2 = 1.0$. Since the stub section is lossless, its input impedance must be purely reactive, i.e. of the form $z_2 = jx$, and hence the input admittance of the loaded section must have the form $z_1 = 1.0 - jx$. The locus of z_1 is thus the unit resistance circle.

The unknown reactive part of z_1 is determined by moving from the point on the chart corresponding to z_L toward the generator until the intersection with the unit resistance circle is reached, yielding $z_1 = 1.0 + j1.13$ and thus $z_2 = -j1.13$. The electrical distance moved between the points z_L and z_1 is $\ell_1 = 0.235\lambda$, the sum of 0.069λ measured on the WAVELENGTHS TOWARD LOAD scale and 0.166λ measured on the WAVELENGTHS TOWARD GENERATOR scale (Why different scales?).

To find ℓ_2 , locate $z_2 = -j1.13$ and move toward the load until the impedance is zero, corresponding to a shorted stub. The shorted stub length is therefore $\ell_2 = 0.365\lambda$, which can be measured directly on the WAVELENGTHS TOWARD GENERATOR scale.

A second solution may be found by rotating past the first intersection to the second intersection with the unit resistance circle, yielding $z_1 = 1.0 - j1.13$ and a longer section of line ℓ_1 . Then $z_2 = j1.13$ and the corresponding length of the shorted stub section ℓ_2 will be shorter than in the previous solution. In this case it is not clear which solution is preferable. In general, however, if one solution results in *both* ℓ_1 and ℓ_2 being shorter, then that configuration normally has the greater bandwidth.





10. Use a single parallel stub tuner to match the above line to its normalized load. Use a shorted stub and find its distance from the load, ℓ_1 , and its length ℓ_2 .

First, locate the load impedance $z_L = 0.3 + j0.5$ on the chart and reflect it to obtain the load admittance, $y_L = 0.89 - j1.48$. To match the line, the parallel combination of the load and stub line sections must present a unit (normalized) admittance at the junction point, i.e. $y_1 + y_2 = 1.0$. Since the stub section is lossless, its input admittance must be purely susceptive, i.e. of the form $y_2 = jb$, and hence the input admittance of the loaded section must have the form $y_1 = 1.0 - jb$. The locus of y_1 is thus the unit conductance circle.

To determine the unknown susceptive part of y_1 , we move from the point on the chart corresponding to y_L toward the generator until we reach the intersection with the unit conductance circle, yielding $y_1 = 1.0 + j1.56$ and thus $y_2 = -j1.56$. The electrical distance moved between the points y_L and y_1 is $\ell_1 = 0.350\lambda$, the sum of 0.172λ measured on the WAVELENGTHS TOWARD LOAD scale and 0.178λ measured on the WAVELENGTHS TOWARD GENERATOR scale (Why different scales?).

To find ℓ_2 , locate $y_2 = -j1.56$ and move toward the load to the infinite admittance point on the chart, corresponding to a shorted stub. The shorted stub length is therefore $\ell_2 = 0.250\lambda - 0.159\lambda = 0.091\lambda$, where both distances are measured on the WAVELENGTHS TOWARD LOAD scale.

As usual, a second solution may be obtained by rotating past the first intersection to the second intersection with the unit conductance circle at $y_1 = 1.0 - j1.56$. This will yield a longer section of line ℓ_1 . Also, we will have $y_2 = j1.56$ with the corresponding length of the shorted stub section ℓ_2 also longer than in the previous solution. In this case one would generally choose the solution with both ℓ_1 and ℓ_2 shorter, that configuration normally having the greater bandwidth.

